

Which Captions Should Be Positive for Image–Text Retrieval?

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Abstract

An image can support several correct descriptions. Training on those descriptions can improve relation accuracy while reducing retrieval accuracy. We separate the effects of caption identity, positive-target allocation, and preservation of the pretrained model. Matched randomized controls keep the candidate captions, positive counts, and initial similarity groups fixed. We test linear and nonlinear adaptation, then repeat the study with a different pretrained encoder. We also combine source-target allocation with source-gallery distillation. Development data selects preservation models under the same retrieval constraint. Both new settings reproduce the image-to-caption cost against source-only training at every tested rate. Caption identity improves caption-to-image retrieval, although the nonlinear evidence varies across rates. At matched schedules chosen for their combination, allocation and distillation each improve retrieval. No selected family meets the joint retrieval and SugarCrep++ criterion in either preservation study. An exact distance to ranking failure explains what distribution preservation can certify on a fixed gallery. These results separate gains from additional supervision from successful retrieval preservation.

1 Introduction

A photograph can support several correct descriptions. Some identify the scene precisely; others fit many images. Which descriptions should retrieval training reward, and how strongly?

Contrastive image–text models reward associated captions relative to competing candidates (Radford et al., 2021). We call these rewarded pairs *positives*. Relation annotations identify supported statements that can supply additional positives. However, promoting a caption changes both its training role and the distribution of supervision. The original captions share their target probabil-

ity with more descriptions. The reverse loss also averages over more caption queries.

Supported descriptions can teach useful visual distinctions without identifying the exact caption assigned to an image. The practical objective combines both capabilities: learn useful distinctions while retaining reliable search.

We study this objective through two linked comparisons. First, we test caption identity while matching positive counts, initial similarity groups, learning rates, and epochs. We repeat the comparison with a nonlinear adapter and a second pretrained encoder. Directional interventions separate image-positive expansion from reverse-anchor expansion. Second, we separate source-target allocation from a standard preservation penalty. Every preservation procedure uses the same development rule and an explicit retrieval tolerance.

Prior methods convert false negatives into positives and modify negative weights (Byun et al., 2024; Jiang et al., 2023). Composition studies already examine valid descriptions and retained capabilities (Kamath et al., 2024; Oh et al., 2024; Peleg et al., 2025). CLIP-Refine combines positive supervision with teacher outputs; WiSE-FT interpolates pretrained and adapted weights (Yamaguchi et al., 2025; Wortsman et al., 2022). Our contribution concerns what caption assignment and allocation add under these preservation controls. Matched allocation and distillation improve retrieval, but neither preservation study yields a family meeting our joint criterion.

2 Assignment and Preservation

A fixed candidate pool. Each training image has five source captions and hypotheses from e-SNLI-VE (Kayser et al., 2021). Hypotheses carry supported, contradicted, or neutral labels. Every batch contains the same images, source captions, and hypotheses across policies. Unpromoted hypotheses

remain competitors. Thus the intervention changes positive assignment without changing the available strings.

Source uses only source captions as positives. *Supported*, denoted U , also promotes supported hypotheses for their annotated image. The *score-stratified* control promotes the same number within each of two frozen-similarity groups. It draws assignments independently three times and keeps each assignment fixed during training.

Two effects of positive expansion. Let s_{ij} score image i against caption j . Let P_i contain its positive captions. The image-query loss is

$$\ell_i = \log \sum_j e^{s_{ij}} - \frac{1}{|P_i|} \sum_{j \in P_i} s_{ij}. \quad (1)$$

With K source captions and M promoted hypotheses, each source target falls from $1/K$ to $1/(K + M)$. The reverse loss averages caption queries against the image candidates. Expansion adds eligible hypothesis queries and reduces the source queries' share of that average. The symmetric loss gives each direction weight $1/2$. Matching promotion counts preserves both changes across supported and randomized assignments. Appendix F.3 proves the resulting source-column gradient identity at fixed scores. We also cross *Source* and *Supported* supervision independently in the two loss directions. The four cells measure each directional intervention and their interaction at matched schedules.

Allocation and distillation as separate interventions. Let L_S and L_U denote the complete *Source* and *Supported* losses. Both losses use the unchanged candidate pool. We compare the family

$$L_{\lambda, \beta} = \lambda L_S + (1 - \lambda) L_U + \beta T^2 D_S. \quad (2)$$

Here λ controls source allocation and β controls preservation. D_S is the mean teacher-to-student KL divergence on source-caption columns in both directions. The frozen encoder supplies the teacher probabilities at temperature $T = 2$. Hypotheses receive supervised gradients but do not enter the distillation target.

The four cells separate the two interventions: U uses $(\lambda, \beta) = (0, 0)$; A uses $(\lambda, 0)$; D uses $(0, \beta)$; AD uses (λ, β) . Allocation mixes complete objectives. For example, the image-row source mass becomes $\lambda + (1 - \lambda)K/(K + M)$.

Fair preservation controls. We also evaluate *Source* and *WiSE-FT* interpolation of *Supported* weights (Wortsman et al., 2022). All procedures receive the same development objective and epoch choices. AD receives a score-stratified randomized control with identical coefficients, learning rate, and selected epochs.

The paired four-cell analysis uses the AD -selected schedule for U , A , D , and AD . For any outcome Y , its interaction is

$$\mathcal{I}_Y = [Y(AD) - Y(D)] - [Y(A) - Y(U)]. \quad (3)$$

An independently selected AD -versus- D comparison instead evaluates complete procedures.

3 Experimental Design

Three adaptation settings. The starting encoder is OpenCLIP ViT-B/32 pretrained on LAION-2B (Cherti et al., 2023). Its original adapters are linear. The nonlinear replication uses separate 128-unit ReLU branches with zero output initialization. The second encoder is OpenAI RN50 with linear adapters (Radford et al., 2021). Its architecture, pretraining, and feature dimension change together. Both encoders remain frozen. Each replication crosses *Source*, *Supported*, and three stratified draws with three rates and three seeds: 45 fits. Rates are 10^{-4} , 3×10^{-4} , and 10^{-3} ; seeds are 17, 29, and 43. Primary comparisons use epoch ten and identical rates.

A separate preservation study. The AD study uses linear adapters and the same 1,200 training images for both encoders. It crosses $\lambda \in \{.5, .8\}$ and $\beta \in \{1, 4, 16\}$ with the declared rates and seeds. *Source*, U , A , D , AD , and nine randomized controls require 126 fits per encoder. Base fits have ten epochs; controls stop at their matched AD epoch. Epoch zero remains available. The earlier 180-fit design separately tests directional interventions and retention. Appendix A gives both full grids, interpolation candidates, and selection rules.

Development selection. Development retrieval uses 900 images and 4,500 source captions, excluding calibration images. Its original 100-image validation subset supplies relation accuracy. Each candidate and seed selects the most accurate epoch that satisfies both retrieval constraints. Each direction can lose at most one percentage point against frozen initialization. Shared coefficients

and a shared rate maximize mean selected accuracy across seeds. Fixed tie rules favor earlier epochs, lower rates, and lower coefficients.

Search and caption discrimination. Image queries retrieve one of five source captions: I→T. Caption queries retrieve their owning image: T→I. R@1 counts queries whose highest-scoring candidate is relevant. The clean e-ViL pool contains 1,000 test images and 5,000 source captions. Both directions search the complete pool. SugarCrepe++ requires both valid captions to outrank an incorrect caption (Dumpala et al., 2024). Secondary endpoints cover SugarCrepe, relation accuracy, and COCO retrieval. Appendix A specifies their pools (Hsieh et al., 2023).

A joint practical criterion. All three selected seeds must contain nonzero updates. Against frozen initialization, both adjusted retrieval bounds must exceed −1 percentage point. The adjusted SugarCrepe++ gain must exceed zero. The same family must pass for both encoders. The AD study adjusts jointly over 80 primary contrasts and separately over 24 factorial contrasts. Both use 100,000 paired image-cluster resamples. Original retention keeps separate 80-effect and 36-effect families; each replication keeps its own twelve-effect family. Caption queries stay with their image during resampling. Intervals condition on fitted seeds, assignment draws, and fixed galleries.

4 Results

The source-retrieval cost recurs across settings. Supported training reduces image-to-caption R@1 against Source at all three rates in both new replications. The nonlinear effects are −3.03, −3.43, −8.37 percentage points across increasing rates. The RN50 effects are −5.00, −6.17, −7.27 points. Every adjusted interval lies below zero. Figure 1 includes the original linear study and both new replications.

Caption identity helps the reverse direction. Against stratified promotion, RN50 gains +2.10, +5.39, +4.63 caption-to-image points. All three adjusted lower bounds exceed zero. The nonlinear gains are +0.94, +0.51, +2.69 points. Its smallest and largest rates have positive lower bounds; the middle interval includes zero. All RN50 image-to-caption intervals against stratified promotion include zero. Nonlinear image-to-caption costs against stratified promotion have negative adjusted

Family	ViT			RN50		
	I→T	T→I	Both	I→T	T→I	Both
Frozen	84.50	69.76	68.95	80.50	59.24	67.35
Source	84.50	68.96	66.76	82.20	62.47	65.02
U	84.50	69.76	68.95	80.50	59.24	67.35
A	83.83	69.03	67.43	81.27	61.69	65.90
D	83.10	70.21	69.31	80.13	61.44	68.26
AD	83.37	68.29	68.82	79.77	59.71	67.97
WiSE	83.80	68.87	69.27	80.30	63.36	67.37
AD random	83.23	68.10	67.72	81.21	59.99	67.86

Table 1: Selected AD-study procedures. Entries are percentages. Both denotes SugarCrepe++ both-caption accuracy. U is frozen for both encoders. RN50 A includes one frozen seed. None of six selected families passes the joint criterion.

intervals at the two smaller rates. Assignment benefits therefore depend on retrieval direction. Appendix Tables 4 and 5 report every new replication effect.

Image-target restriction improves image-to-caption retrieval. Restricting image targets to source captions improves image-to-caption retrieval at every tested rate for both encoders. The averaged effects are 5.83, 6.28, and 9.52 points for ViT. RN50 gains 4.92, 5.68, and 7.27 points. All six adjusted intervals exclude zero. Reverse-anchor effects on this endpoint and all directional interactions include zero. Caption-to-image effects vary across rates and encoders. These comparisons isolate the defined directional intervention, not target dilution as its sole mechanism.

No selected procedure meets the practical criterion. Neither study produces a passing family on either encoder: seven original-retention families and six AD-study families. AD produces nonzero updates for all six selected seeds. Its relation-accuracy gains over frozen initialization are +6.67 and +9.02 points for ViT and RN50. Both adjusted intervals exclude zero. However, its SugarCrepe++ changes are −0.13 and +0.62 points; both intervals include zero. AD’s ViT caption-to-image loss is 1.47 points, with an adjusted interval below zero. The other three AD retrieval intervals include zero but extend below the one-point tolerance. Those intervals do not establish retention or prove a large retrieval loss. Thus relation learning does not establish the requested caption improvement with retained retrieval. Appendix Table 2 reports every criterion and lower bound. The original-retention retrieval intervals show no established decline against frozen initialization. How-

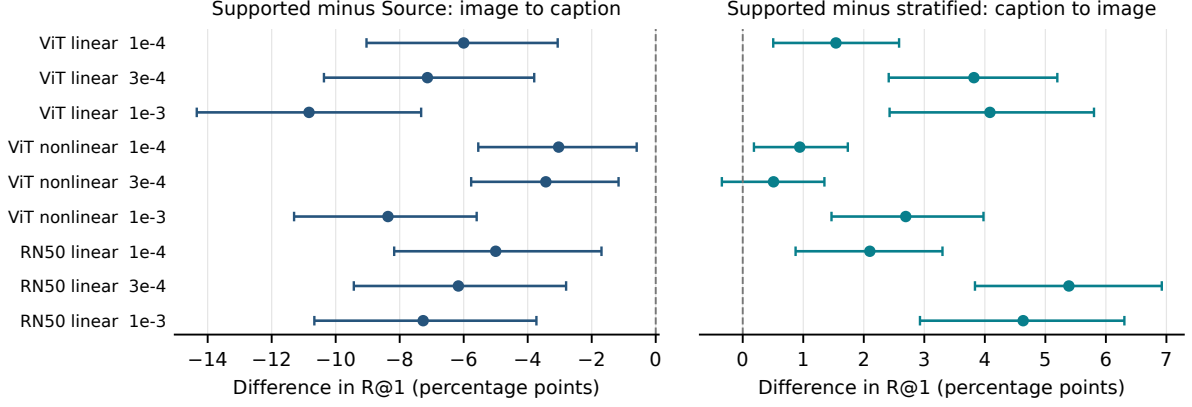


Figure 1: Caption identity and source retrieval at matched epoch ten. Dots average training seeds and fixed assignment draws. Bars show the adjusted intervals from each setting’s separately declared comparison family. The appendix reports all retrieval directions and comparators.

ever, Source, A, and image-source-only training reduce SugarCrepe++ accuracy for both encoders. U and reverse-source-only select frozen initialization throughout. Appendix Table 3 retains all original-retention failures.

Matched effects differ from practical success.

At the AD-selected schedule, allocation and distillation each improve both retrieval directions on both encoders. All eight adjusted main-effect intervals lie above zero. The distillation main effect also improves SugarCrepe++ by +2.17 and +3.43 points. These contrasts compare matched objective cells, rather than each selected procedure with frozen initialization. All retrieval and SugarCrepe++ interaction intervals include zero. Allocation reduces relation accuracy at these matched schedules. RN50 has a negative relation-accuracy interaction. No positive interaction is established. Separately selected AD loses 1.93 and 1.73 caption-to-image points against D; both adjusted intervals exclude zero.

5 What Preservation Can Certify

Distillation controls a probability distribution, whereas retrieval depends on the highest-ranked relevant candidate. The following result connects them on a fixed gallery. Let q and p denote teacher and student probabilities on that gallery. Let R contain its relevant candidates. Define $C_K(q, R)$ as the smallest forward KL divergence to ranking failure. Failure makes K irrelevant candidates tie or exceed every relevant candidate.

Proposition 1. Assume the teacher is strictly correct and at least K irrelevant candidates exist. Se-

lect the K largest irrelevant teacher probabilities. Cap relevant probabilities above a level t and raise selected irrelevant probabilities below t . Choose t so the removed and added mass agree. Call the resulting distribution p^* . Then $C_K(q, R) = \text{KL}(q||p^*)$. Consequently, $\text{KL}(q||p) < C_K(q, R)$ guarantees a relevant candidate within the first K positions.

Appendix D gives the formula, attaining projection, proof, and numerical boundary cases. The radius treats all valid candidates jointly. It can exceed a bound that protects only the teacher’s best relevant candidate. The same-gallery requirement matters: a minibatch penalty cannot certify unseen distractors in a larger gallery.

Appendix C reports audited full-gallery coverage separately for trained and unchanged selections. Coverage varies by encoder, retrieval direction, and gallery.

6 Conclusion

Caption identity can help caption-to-image retrieval, while expanded supervision reduces image-to-caption retrieval against source-only training. Image-target restriction consistently improves the latter endpoint across both encoders. Allocation and distillation improve matched cells, but practical retrieval preservation with better caption discrimination remains unresolved.

Limitations

The experiments use frozen encoders, small residual adapters, and 1,200 training images. They do not establish behavior under full-encoder training or larger data scales. Changing to RN50 changes architecture, pretraining, and feature dimension together. The replication does not isolate those factors.

Some supported labels inherit text-inference judgments rather than independent visual adjudication. Source-caption ownership also provides incomplete retrieval relevance. The score-stratified controls match coarse similarity groups rather than exact scores or all semantic properties. Their quotas depend on relation annotations.

The test sets informed earlier stages and remain reused evaluation sets. The reconstructed experiments and AD extension remain exploratory relative to that prior work. The development pool includes the original validation subset. It excludes calibration images and does not overlap the test pool. The bootstrap conditions on fitted seeds, assignment draws, and fixed candidate galleries. It does not resample training runs or construct new galleries. Failure to establish one-point noninferiority does not prove that the true retrieval loss exceeds one point. The adjusted intervals report that uncertainty.

The historical linear study retains its manuscript, aggregate results, and original audit receipts. Its raw prediction archives are unavailable in the recovered workspace. The current 522 executions have separate, newly saved evaluation artifacts and independent statistical audits. They do not reconstruct the missing historical predictions.

The rank certificate applies to identical queries, galleries, relevance sets, and probability temperatures. Its proof does not transfer a minibatch training penalty to unseen queries. Frozen-teacher distillation, weight interpolation, and convex objective mixtures are established ingredients. The controlled comparisons determine their value in this setting.

Ethical Considerations and Assistance

We use existing public datasets under their source terms and collect no new human judgments. Captions and pretrained features can reflect source biases. Generative AI assisted planning, implementation, analysis, literature discovery, and drafting.

The authors are responsible for checking the released computations and manuscript.

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A Replication and Preservation: Complete Methods

A.1 Study boundaries

The starting study contains 72 matched-assignment fits. Its full methods and results appear in Appendix E. The nonlinear and RN50 replications each contain 45 fits. Their protocols are `docs/STRENGTHEN_REPLICATION_PROTOCOL.json` and `docs/STRENGTHEN_ENCODER_REPLICATION_PROTOCOL.json`. The frozen retention design contributes 180 fits under `results/strengthen_retention/protocol_v3.json`. The preservation extension has its own protocol, `results/allocation_distillation/protocol_v1.json`. This extension adds the allocation-plus-distillation comparison to the earlier design. It uses previously examined test sets and is identified as exploratory. The scientific record keeps every earlier protocol separately. Some Source, U, A, and D configurations recur

across these stages. Identical inputs, seeds, objectives, and schedules define repeated executions of the same seeded fit. Execution totals therefore do not increase the number of independent training seeds. Each inference family uses only its declared seed and assignment sets.

New strengthening results require available prediction archives and completed analysis receipts. Earlier reported strengthening outputs do not supply numbers for this revision. The repository recovery ledger records that execution boundary.

A.2 The frozen directional and retention design

The retention design compares seven selected families for each encoder. They are Source, Supported, allocation, distillation, WiSE-FT, image-source-only, and reverse-source-only. The underlying grid has nine configurations, three rates, and three seeds. This gives 81 base fits per encoder. Nine matched randomized distillation controls raise each encoder’s total to 90. These controls copy D’s primary selected coefficient, rate, and per-seed epoch. They do not copy the later AD schedule.

Write L_I^S, L_T^S for the source image and caption losses. Write L_I^U, L_T^U for the corresponding expanded losses. The four directional cells are

$$S = \frac{1}{2}(L_I^S + L_T^S), \quad U = \frac{1}{2}(L_I^U + L_T^U), \quad (4)$$

$$I_S = \frac{1}{2}(L_I^S + L_T^U), \quad T_S = \frac{1}{2}(L_I^U + L_T^S). \quad (5)$$

All four retain the same candidate captions. I_S restricts the image targets to source captions. T_S restricts eligible reverse anchors to source captions. The reverse intervention changes which captions participate in the average. It is not solely a change to individual image-target probabilities.

For outcome Y , the source-restriction main effects are

$$\Delta_I = \frac{1}{2}[Y(S) + Y(I_S) - Y(U) - Y(T_S)], \quad (6)$$

$$\Delta_T = \frac{1}{2}[Y(S) + Y(T_S) - Y(U) - Y(I_S)]. \quad (7)$$

The interaction is $Y(S) - Y(I_S) - Y(T_S) + Y(U)$. These contrasts use epoch ten at each of the three rates. Three contrasts, two retrieval endpoints, three rates, and two encoders give 36 effects. Their 100,000 bootstrap resamples use seed 20261006. The adjustment covers this complete directional family.

The separate selected-procedure family contains 80 contrasts. Seven families against frozen contribute 28 effects per encoder across four endpoints. D against A and WiSE-FT contributes

eight. D against matched randomized distillation contributes four. The adjustment covers both encoders jointly. Its 100,000 bootstrap resamples use seed 20261005.

Selection, coefficients, practical thresholds, and zero-loss sensitivity follow the unchanged frozen protocol. The later AD extension adds its missing combined cell under a separate protocol. Its factorial concerns allocation and distillation, not the image and reverse directional interventions. All inference families remain distinct.

A.3 Models and features

ViT-B/32 uses the OpenCLIP LAION-2B checkpoint and its native score scale. Its image and text features have 512 dimensions. RN50 uses the official OpenAI checkpoint and its own preprocessing and score scale. Its features have 1,024 dimensions. Both encoders remain fixed during adaptation.

For a normalized feature x , a linear residual adapter returns

$$f_W(x) = \frac{x + Wx}{\|x + Wx\|_2}. \quad (8)$$

The two modalities have independent matrices. Both matrices start at zero. The ViT adapter has 524,288 parameters; the RN50 adapter has 2,097,152 parameters. No projection bias is trained.

The nonlinear ViT replication returns

$$f_{A,B}(x) = \frac{x + B \operatorname{ReLU}(Ax)}{\|x + B \operatorname{ReLU}(Ax)\|_2}. \quad (9)$$

Each modality uses a 128-dimensional hidden layer. The output matrix B starts at zero. The input matrix can learn after the output matrix changes. The two branches contain 262,144 trainable parameters in total. This replication changes adaptation capacity while retaining the same pretrained representation.

Every feature archive carries the encoder identity, checkpoint hash, preprocessing, image order, and text order. Recomputed feature archives remain identified as fresh executions. They are not described as exact-byte recoveries.

A.4 Candidate and assignment invariants

Training uses 1,200 images with their five source captions and all retained hypotheses. Each hypothesis belongs to its annotated image. Both replicated policies and random controls use the same complete batch candidates. Unpromoted hypotheses remain competitors.

The score-stratified control ranks each image’s hypotheses by its own encoder’s frozen cosine scores. Manifest index resolves exact score ties. The ordered list splits into two nearly equal bins. Each bin receives exactly its original supported count. Sampling uses NumPy PCG64 seeds 101, 211, and 307. Each assignment remains fixed across all rates, training seeds, epochs, and batches. Naturally unchanged assignments remain in the sample. The controls retain the original labels for evaluation and provenance. Appendix F.2 gives the full assignment definition.

A.5 Exact preservation objective

Let Z contain the scaled image–text scores for a batch. Let C contain only its source-caption columns. Teacher and student row distributions are

$$q_i^I = \operatorname{softmax}(Z_{i,C}^0/T), \quad (10)$$

$$p_i^I = \operatorname{softmax}(Z_{i,C}/T). \quad (11)$$

Column distributions q_j^T, p_j^T normalize the same source submatrix over image rows. The preservation term is

$$D_S = \frac{1}{2B} \sum_{i=1}^B \operatorname{KL}(q_i^I \| p_i^I) + \frac{1}{2|C|} \sum_{j \in C} \operatorname{KL}(q_j^T \| p_j^T). \quad (12)$$

Teacher probabilities do not receive gradients. The temperature is $T = 2$. Equation 2 includes the multiplier T^2 outside this divergence. Thus $\beta \in \{1, 4, 16\}$ gives effective multipliers 4, 16, and 64.

L_S and L_U each average image and eligible-caption losses with equal directional weight. The mixture $\lambda L_S + (1 - \lambda) L_U$ therefore mixes complete symmetric objectives. It does not simply redistribute image-row targets while leaving the reverse loss unchanged. For K source captions and M supports, the row source mass is

$$\lambda + (1 - \lambda) \frac{K}{K + M}. \quad (13)$$

The reverse source-anchor contribution also increases with λ . Its precise mass depends on the current batch’s eligible anchor counts.

A.6 Complete fit grid

Each encoder has 13 deterministic configurations. Source and U contribute one each. A contributes two allocation coefficients. D contributes three

distillation coefficients. AD contributes all six coefficient pairs. Three learning rates and three seeds produce 117 fits per encoder. Nine matched randomized AD controls raise that count to 126. The full extension therefore contains 252 fits.

Base fits use ten epochs, batches of 32 images, and the same seed-specific image order. The rates are 10^{-4} , 3×10^{-4} , and 10^{-3} . The seeds are 17, 29, and 43. AdamW uses weight decay .01, coefficients (.9, .999), and epsilon 10^{-8} . Gradient norms are clipped at one. There is no learning-rate scheduler. Deterministic CPU fitting uses two threads. The full protocol records implementation and feature hashes before fitting.

WiSE-FT interpolates each Supported correction matrix with its zero initialization. The candidate coefficients are $\alpha \in \{0, .1, .25, .5, .75, 1\}$. The same α applies to both modalities. Interpolation uses model parameters before feature normalization. It does not average normalized features or final scores.

A.7 Selection rules

The development retrieval pool has 900 images and 4,500 source captions. It excludes the original 100 calibration images. It includes the original 100-image validation subset used for relation accuracy. This overlap is intentional and does not create a second independent development sample.

For each candidate and seed, an epoch is feasible when both development retrieval constraints hold. The primary tolerance permits a one-percentage-point loss from each frozen reference. Epoch zero is always available. Among feasible epochs, selection maximizes image-balanced supported-versus-contradicted accuracy. Relation ties receive half credit. Exact ties retain the earlier epoch.

Each family then selects shared coefficients and a shared rate across the three seeds. Selection maximizes the mean chosen relation score. The protocol fixes the remaining tie order. The zero-percentage-point tolerance repeats the procedure as a descriptive sensitivity analysis. It does not introduce additional primary hypotheses.

Matched randomized AD controls copy the primary AD coefficients, rate, and per-seed epoch. They use three assignment draws crossed with three training seeds. They are fitted without an independent development search. The fitted control run stops at the required selected epoch. An epoch-zero match remains explicitly frozen.

A.8 Independent selection and matched decomposition

The primary comparisons evaluate independently selected Source, U, A, D, AD, and WiSE-FT procedures. The matched random comparison evaluates caption identity under the primary AD schedule. The decomposition instead takes all four cells at that AD-selected schedule. It uses the selected allocation coefficient for A and AD. It uses the selected distillation coefficient for D and AD.

For outcome Y , let $Y_{00}, Y_{10}, Y_{01}, Y_{11}$ denote U, A, D, and AD respectively. The conventional allocation effect is

$$\Delta_A = \frac{1}{2}[(Y_{10} - Y_{00}) + (Y_{11} - Y_{01})]. \quad (14)$$

The conventional distillation effect is

$$\Delta_D = \frac{1}{2}[(Y_{01} - Y_{00}) + (Y_{11} - Y_{10})]. \quad (15)$$

The interaction is

$$\Delta_{AD} = Y_{11} - Y_{01} - Y_{10} + Y_{00}. \quad (16)$$

The interaction measures departure from additivity for this outcome and selected schedule. It does not establish a universal mechanism across objectives or training distributions.

A.9 Evaluation and inference

Primary endpoints are e-ViL $I \rightarrow T$ R@1, e-ViL $T \rightarrow I$ R@1, relation accuracy, and SugarCrep++ both-caption accuracy. The preservation family contains 80 contrasts across two encoders. The decomposition contains 24 contrasts: three factorial effects, four endpoints, and two encoders. Each family uses 100,000 paired image-cluster bootstrap resamples and its declared Bonferroni adjustment. The bootstrap retains each image’s associated caption queries and benchmark records. Randomized controls average assignment draws within each training seed before seed averaging. The fixed seeds and draws are not themselves resampled.

The practical gate uses adjusted lower bounds against frozen initialization. Both retrieval bounds must exceed -1 percentage point. The SugarCrep++ bound must exceed zero. Each selected state must have a positive epoch and nonzero update norm. An across-encoder success requires the same family to pass for both encoders. Point estimates alone cannot establish the gate. Retrieval intervals can cross zero if their lower bounds exceed the declared negative tolerance.

The replication families remain separate from preservation inference. Each has 12 contrasts at epoch ten. They compare Supported with Source and score-stratified promotion at every rate in both retrieval directions. The original linear study retains its separate 18-contrast family.

Current strengthening predictions retain query identities and aggregate definitions. Independent audits recompute reported effects from those predictions. Source image overlap, checkpoint selection, and protocol hashes receive separate checks. The complete tables below report every declared comparison.

B Replication and Preservation: Complete Results

All differences below use percentage points.

Intervals use the separately declared comparison families.

A separate audit verified 144 pairwise comparisons of repeated fits and 1,584 matched trajectory rows.

The terminal parameter tensors agree in every comparison.

These deterministic duplicate executions do not add independent training seeds.

Encoder	Family	Trained	I→T lower	T→I lower	Both lower	Joint
ViT	Source	Yes	-2.800	-2.338	-3.502	Fail
ViT	U	No	+0.000	+0.000	+0.000	Fail
ViT	A	Yes	-3.533	-2.225	-2.779	Fail
ViT	D	Yes	-3.933	-0.512	-0.613	Fail
ViT	AD	Yes	-4.067	-2.913	-1.456	Fail
ViT	WiSE	Yes	-3.267	-2.438	-0.923	Fail
RN50	Source	Yes	-1.867	+1.195	-4.035	Fail
RN50	U	No	+0.000	+0.000	+0.000	Fail
RN50	A	No	-1.667	+1.027	-2.660	Fail
RN50	D	Yes	-3.733	+0.653	-0.613	Fail
RN50	AD	Yes	-4.000	-1.207	-0.901	Fail
RN50	WiSE	Yes	-4.000	+2.135	-1.556	Fail

Table 2: AD-study practical criteria. Trained requires all three nonzero selected updates. Numbers are adjusted lower bounds against frozen initialization, in percentage points. Both retrieval bounds must exceed -1 , and the Both bound must exceed zero.

Encoder	Family	Trained	I→T lower	T→I lower	Both lower	Joint
ViT	Source	Yes	-2.833	-2.333	-3.567	Fail
ViT	U	No	+0.000	+0.000	+0.000	Fail
ViT	A	Yes	-3.567	-2.187	-2.798	Fail
ViT	D	Yes	-3.867	-0.487	-0.578	Fail
ViT	WiSE	Yes	-3.192	-2.373	-0.880	Fail
ViT	Image source	Yes	-2.500	-2.265	-3.599	Fail
ViT	Reverse source	No	+0.000	+0.000	+0.000	Fail
RN50	Source	Yes	-1.767	+1.183	-4.034	Fail
RN50	U	No	+0.000	+0.000	+0.000	Fail
RN50	A	No	-1.633	+1.015	-2.617	Fail
RN50	D	Yes	-3.767	+0.660	-0.654	Fail
RN50	WiSE	Yes	-4.033	+2.183	-1.539	Fail
RN50	Image source	Yes	-1.367	+1.487	-4.053	Fail
RN50	Reverse source	No	+0.000	+0.000	+0.000	Fail

Table 3: Original retention practical criteria use their original jointly adjusted family. Trained requires all three nonzero selected updates. Numbers are adjusted lower bounds against frozen initialization, in percentage points. Both retrieval bounds must exceed -1 , and the Both bound must exceed zero.

Rate	Comparator	Endpoint	Difference	Adjusted interval
0.0001	Source	i2t.r1	-3.033	$[-5.545, -0.594]$
0.0001	Source	t2i.r1	+1.233	$[+0.240, +2.241]$
0.0001	Stratified	i2t.r1	-2.278	$[-4.082, -0.511]$
0.0001	Stratified	t2i.r1	+0.942	$[+0.185, +1.736]$
0.0003	Source	i2t.r1	-3.433	$[-5.767, -1.161]$
0.0003	Source	t2i.r1	-0.207	$[-1.313, +0.908]$
0.0003	Stratified	i2t.r1	-2.678	$[-4.624, -0.878]$
0.0003	Stratified	t2i.r1	+0.509	$[-0.345, +1.350]$
0.001	Source	i2t.r1	-8.367	$[-11.300, -5.594]$
0.001	Source	t2i.r1	-2.160	$[-3.562, -0.740]$
0.001	Stratified	i2t.r1	-0.067	$[-2.485, +2.239]$
0.001	Stratified	t2i.r1	+2.693	$[+1.467, +3.978]$

Table 4: All twelve Supported-minus-comparator effects for ViT nonlinear. Every comparison uses epoch ten and a common learning rate.

Rate	Comparator	Endpoint	Difference	Adjusted interval
0.0001	Source	i2t.r1	-5.000	$[-8.172, -1.694]$
0.0001	Source	t2i.r1	-0.307	$[-1.787, +1.114]$
0.0001	Stratified	i2t.r1	-0.933	$[-3.380, +1.435]$
0.0001	Stratified	t2i.r1	+2.100	$[+0.873, +3.300]$
0.0003	Source	i2t.r1	-6.167	$[-9.433, -2.800]$
0.0003	Source	t2i.r1	-0.660	$[-2.360, +1.034]$
0.0003	Stratified	i2t.r1	+1.589	$[-1.156, +4.369]$
0.0003	Stratified	t2i.r1	+5.389	$[+3.837, +6.924]$
0.001	Source	i2t.r1	-7.267	$[-10.667, -3.728]$
0.001	Source	t2i.r1	-1.173	$[-3.110, +0.739]$
0.001	Stratified	i2t.r1	+1.656	$[-1.180, +4.478]$
0.001	Stratified	t2i.r1	+4.633	$[+2.928, +6.306]$

Table 5: All twelve Supported-minus-comparator effects for RN50 linear. Every comparison uses epoch ten and a common learning rate.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
ViT	Source – Frozen	I→T	+0.000	$[-2.800, +2.858]$
ViT	Source – Frozen	T→I	-0.800	$[-2.338, +0.745]$
ViT	Source – Frozen	Relation	+1.562	$[+0.289, +2.814]$
ViT	Source – Frozen	Both	-2.193	$[-3.502, -0.859]$
ViT	U – Frozen	I→T	+0.000	$[+0.000, +0.000]$
ViT	U – Frozen	T→I	+0.000	$[+0.000, +0.000]$
ViT	U – Frozen	Relation	+0.000	$[+0.000, +0.000]$
ViT	U – Frozen	Both	+0.000	$[+0.000, +0.000]$
ViT	A – Frozen	I→T	-0.667	$[-3.533, +2.233]$
ViT	A – Frozen	T→I	-0.727	$[-2.225, +0.753]$
ViT	A – Frozen	Relation	+4.744	$[+3.549, +6.008]$
ViT	A – Frozen	Both	-1.521	$[-2.779, -0.254]$
ViT	D – Frozen	I→T	-1.400	$[-3.933, +1.000]$
ViT	D – Frozen	T→I	+0.453	$[-0.512, +1.407]$
ViT	D – Frozen	Relation	+5.517	$[+4.249, +6.797]$
ViT	D – Frozen	Both	+0.357	$[-0.613, +1.325]$

Table 6: Primary preservation contrasts, rows 1 through 16.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
ViT	AD – Frozen	I→T	-1.133	$[-4.067, +1.858]$
ViT	AD – Frozen	T→I	-1.473	$[-2.913, -0.017]$
ViT	AD – Frozen	Relation	+6.674	$[+5.210, +8.167]$
ViT	AD – Frozen	Both	-0.126	$[-1.456, +1.196]$
ViT	WiSE – Frozen	I→T	-0.700	$[-3.267, +1.892]$
ViT	WiSE – Frozen	T→I	-0.893	$[-2.438, +0.567]$
ViT	WiSE – Frozen	Relation	+5.084	$[+3.963, +6.241]$
ViT	WiSE – Frozen	Both	+0.315	$[-0.923, +1.537]$
ViT	AD – D	I→T	+0.267	$[-1.700, +2.233]$
ViT	AD – D	T→I	-1.927	$[-3.038, -0.833]$
ViT	AD – D	Relation	+1.157	$[+0.390, +1.948]$
ViT	AD – D	Both	-0.483	$[-1.527, +0.555]$
ViT	AD – A	I→T	-0.467	$[-2.792, +1.925]$
ViT	AD – A	T→I	-0.747	$[-2.167, +0.700]$
ViT	AD – A	Relation	+1.930	$[+0.956, +2.906]$
ViT	AD – A	Both	+1.394	$[+0.204, +2.622]$

Table 7: Primary preservation contrasts, rows 17 through 32.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
ViT	AD – WiSE	I→T	-0.433	[-2.267, +1.433]
ViT	AD – WiSE	T→I	-0.580	[-1.587, +0.413]
ViT	AD – WiSE	Relation	+1.590	[+0.811, +2.403]
ViT	AD – WiSE	Both	-0.441	[-1.304, +0.391]
ViT	AD – AD random	I→T	+0.133	[-1.467, +1.711]
ViT	AD – AD random	T→I	+0.191	[-0.601, +0.975]
ViT	AD – AD random	Relation	+4.272	[+3.440, +5.167]
ViT	AD – AD random	Both	+1.100	[+0.284, +1.928]
RN50	Source – Frozen	I→T	+1.700	[-1.867, +5.233]
RN50	Source – Frozen	T→I	+3.227	[+1.195, +5.283]
RN50	Source – Frozen	Relation	+1.071	[-0.068, +2.218]
RN50	Source – Frozen	Both	-2.333	[-4.035, -0.662]
RN50	U – Frozen	I→T	+0.000	[+0.000, +0.000]
RN50	U – Frozen	T→I	+0.000	[+0.000, +0.000]
RN50	U – Frozen	Relation	+0.000	[+0.000, +0.000]
RN50	U – Frozen	Both	+0.000	[+0.000, +0.000]

Table 8: Primary preservation contrasts, rows 33 through 48.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
RN50	A – Frozen	I→T	+0.767	[-1.667, +3.233]
RN50	A – Frozen	T→I	+2.453	[+1.027, +3.872]
RN50	A – Frozen	Relation	+3.141	[+2.243, +4.097]
RN50	A – Frozen	Both	-1.450	[-2.660, -0.270]
RN50	D – Frozen	I→T	-0.367	[-3.733, +2.967]
RN50	D – Frozen	T→I	+2.200	[+0.653, +3.815]
RN50	D – Frozen	Relation	+10.351	[+8.283, +12.487]
RN50	D – Frozen	Both	+0.904	[-0.613, +2.431]
RN50	AD – Frozen	I→T	-0.733	[-4.000, +2.400]
RN50	AD – Frozen	T→I	+0.467	[-1.207, +2.133]
RN50	AD – Frozen	Relation	+9.016	[+7.145, +10.957]
RN50	AD – Frozen	Both	+0.617	[-0.901, +2.100]
RN50	WiSE – Frozen	I→T	-0.200	[-4.000, +3.533]
RN50	WiSE – Frozen	T→I	+4.120	[+2.135, +6.130]
RN50	WiSE – Frozen	Relation	+9.171	[+7.518, +10.930]
RN50	WiSE – Frozen	Both	+0.021	[-1.556, +1.589]

Table 9: Primary preservation contrasts, rows 49 through 64.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
RN50	AD – D	I→T	-0.367	[-2.400, +1.700]
RN50	AD – D	T→I	-1.733	[-2.893, -0.587]
RN50	AD – D	Relation	-1.334	[-2.070, -0.621]
RN50	AD – D	Both	-0.287	[-1.184, +0.599]
RN50	AD – A	I→T	-1.500	[-4.100, +1.133]
RN50	AD – A	T→I	-1.987	[-3.518, -0.500]
RN50	AD – A	Relation	+5.876	[+4.527, +7.289]
RN50	AD – A	Both	+2.067	[+0.856, +3.286]
RN50	AD – WiSE	I→T	-0.533	[-3.200, +2.233]
RN50	AD – WiSE	T→I	-3.653	[-5.213, -2.093]
RN50	AD – WiSE	Relation	-0.155	[-1.050, +0.734]
RN50	AD – WiSE	Both	+0.596	[-0.463, +1.630]
RN50	AD – AD random	I→T	-1.444	[-3.056, +0.078]
RN50	AD – AD random	T→I	-0.284	[-1.002, +0.466]
RN50	AD – AD random	Relation	+4.266	[+3.414, +5.171]
RN50	AD – AD random	Both	+0.107	[-0.650, +0.850]

Table 10: Primary preservation contrasts, rows 65 through 80.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
ViT	Allocation	I→T	+3.383	[+2.000, +4.817]
ViT	Distillation	I→T	+3.650	[+1.450, +5.833]
ViT	Interaction	I→T	-0.300	[-2.267, +1.667]
ViT	Allocation	T→I	+0.787	[+0.253, +1.320]
ViT	Distillation	T→I	+4.740	[+3.430, +6.096]
ViT	Interaction	T→I	-0.333	[-1.140, +0.460]
ViT	Allocation	Relation	-1.047	[-1.507, -0.619]
ViT	Distillation	Relation	+0.080	[-0.632, +0.789]
ViT	Interaction	Relation	-0.153	[-0.648, +0.365]
ViT	Allocation	Both	-0.543	[-1.133, +0.025]
ViT	Distillation	Both	+2.169	[+1.179, +3.170]
ViT	Interaction	Both	+0.455	[-0.385, +1.308]
RN50	Allocation	I→T	+3.633	[+2.117, +5.183]
RN50	Distillation	I→T	+7.000	[+4.200, +9.817]
RN50	Interaction	I→T	-0.733	[-3.233, +1.733]
RN50	Allocation	T→I	+0.913	[+0.267, +1.583]

Table 11: AD-selected factorial contrasts, rows 1 through 16.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
RN50	Distillation	T→I	+3.253	[+1.400, +5.100]
RN50	Interaction	T→I	-0.720	[-1.860, +0.427]
RN50	Allocation	Relation	-1.256	[-1.707, -0.814]
RN50	Distillation	Relation	-0.618	[-1.435, +0.186]
RN50	Interaction	Relation	-0.850	[-1.507, -0.211]
RN50	Allocation	Both	-0.266	[-0.835, +0.290]
RN50	Distillation	Both	+3.427	[+2.260, +4.586]
RN50	Interaction	Both	-0.322	[-1.204, +0.569]

Table 12: AD-selected factorial contrasts, rows 17 through 24.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
ViT	Source – Frozen	I→T	+0.000	[-2.833, +2.792]
ViT	Source – Frozen	T→I	-0.800	[-2.333, +0.713]
ViT	Source – Frozen	Relation	+1.562	[+0.341, +2.795]
ViT	Source – Frozen	Both	-2.193	[-3.567, -0.891]
ViT	U – Frozen	I→T	+0.000	[+0.000, +0.000]
ViT	U – Frozen	T→I	+0.000	[+0.000, +0.000]
ViT	U – Frozen	Relation	+0.000	[+0.000, +0.000]
ViT	U – Frozen	Both	+0.000	[+0.000, +0.000]
ViT	A – Frozen	I→T	-0.667	[-3.567, +2.133]
ViT	A – Frozen	T→I	-0.727	[-2.187, +0.753]
ViT	A – Frozen	Relation	+4.744	[+3.557, +6.002]
ViT	A – Frozen	Both	-1.521	[-2.798, -0.271]
ViT	D – Frozen	I→T	-1.400	[-3.867, +1.100]
ViT	D – Frozen	T→I	+0.453	[-0.487, +1.380]
ViT	D – Frozen	Relation	+5.517	[+4.259, +6.884]
ViT	D – Frozen	Both	+0.357	[-0.578, +1.337]

Table 13: Original retention contrasts, rows 1 through 16.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
ViT	WiSE – Frozen	I→T	-0.700	[-3.192, +1.767]
ViT	WiSE – Frozen	T→I	-0.893	[-2.373, +0.613]
ViT	WiSE – Frozen	Relation	+5.084	[+3.955, +6.260]
ViT	WiSE – Frozen	Both	+0.315	[-0.880, +1.550]
ViT	Image source – Frozen	I→T	+0.333	[-2.500, +3.100]
ViT	Image source – Frozen	T→I	-0.560	[-2.265, +1.092]
ViT	Image source – Frozen	Relation	+1.582	[+0.333, +2.815]
ViT	Image source – Frozen	Both	-2.214	[-3.599, -0.880]
ViT	Reverse source – Frozen	I→T	+0.000	[+0.000, +0.000]
ViT	Reverse source – Frozen	T→I	+0.000	[+0.000, +0.000]
ViT	Reverse source – Frozen	Relation	+0.000	[+0.000, +0.000]
ViT	Reverse source – Frozen	Both	+0.000	[+0.000, +0.000]
ViT	D – A	I→T	-0.733	[-3.267, +1.833]
ViT	D – A	T→I	+1.180	[+0.002, +2.365]
ViT	D – A	Relation	+0.773	[-0.262, +1.804]
ViT	D – A	Both	+1.878	[+0.719, +3.071]

Table 14: Original retention contrasts, rows 17 through 32.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
ViT	D – WiSE	I→T	-0.700	[-2.567, +1.233]
ViT	D – WiSE	T→I	+1.347	[+0.213, +2.460]
ViT	D – WiSE	Relation	+0.433	[-0.394, +1.282]
ViT	D – WiSE	Both	+0.042	[-0.927, +0.997]
ViT	D – D random	I→T	-0.233	[-1.678, +1.167]
ViT	D – D random	T→I	-0.198	[-0.658, +0.240]
ViT	D – D random	Relation	+2.853	[+2.211, +3.554]
ViT	D – D random	Both	+0.605	[-0.016, +1.246]
RN50	Source – Frozen	I→T	+1.700	[-1.767, +5.200]
RN50	Source – Frozen	T→I	+3.227	[+1.183, +5.352]
RN50	Source – Frozen	Relation	+1.071	[-0.064, +2.250]
RN50	Source – Frozen	Both	-2.333	[-4.034, -0.712]
RN50	U – Frozen	I→T	+0.000	[+0.000, +0.000]
RN50	U – Frozen	T→I	+0.000	[+0.000, +0.000]
RN50	U – Frozen	Relation	+0.000	[+0.000, +0.000]
RN50	U – Frozen	Both	+0.000	[+0.000, +0.000]

Table 15: Original retention contrasts, rows 33 through 48.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
RN50	A – Frozen	I→T	+0.767	[-1.633, +3.225]
RN50	A – Frozen	T→I	+2.453	[+1.015, +3.938]
RN50	A – Frozen	Relation	+3.141	[+2.250, +4.093]
RN50	A – Frozen	Both	-1.450	[-2.617, -0.329]
RN50	D – Frozen	I→T	-0.367	[-3.767, +3.033]
RN50	D – Frozen	T→I	+2.200	[+0.660, +3.773]
RN50	D – Frozen	Relation	+10.351	[+8.257, +12.513]
RN50	D – Frozen	Both	+0.904	[-0.654, +2.443]
RN50	WiSE – Frozen	I→T	-0.200	[-4.033, +3.533]
RN50	WiSE – Frozen	T→I	+4.120	[+2.183, +6.118]
RN50	WiSE – Frozen	Relation	+9.171	[+7.523, +10.921]
RN50	WiSE – Frozen	Both	+0.021	[-1.539, +1.562]
RN50	Image source – Frozen	I→T	+1.900	[-1.367, +5.333]
RN50	Image source – Frozen	T→I	+3.607	[+1.487, +5.777]
RN50	Image source – Frozen	Relation	+1.348	[+0.224, +2.539]
RN50	Image source – Frozen	Both	-2.319	[-4.053, -0.617]

Table 16: Original retention contrasts, rows 49 through 64.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
RN50	Reverse source – Frozen	I→T	+0.000	[+0.000, +0.000]
RN50	Reverse source – Frozen	T→I	+0.000	[+0.000, +0.000]
RN50	Reverse source – Frozen	Relation	+0.000	[+0.000, +0.000]
RN50	Reverse source – Frozen	Both	+0.000	[+0.000, +0.000]
RN50	D – A	I→T	-1.133	[-3.833, +1.633]
RN50	D – A	T→I	-0.253	[-1.620, +1.092]
RN50	D – A	Relation	+7.210	[+5.622, +8.864]
RN50	D – A	Both	+2.354	[+1.070, +3.629]
RN50	D – WiSE	I→T	-0.167	[-2.667, +2.333]
RN50	D – WiSE	T→I	-1.920	[-3.253, -0.593]
RN50	D – WiSE	Relation	+1.179	[+0.156, +2.179]
RN50	D – WiSE	Both	+0.883	[-0.079, +1.864]
RN50	D – D random	I→T	-0.733	[-2.822, +1.375]
RN50	D – D random	T→I	-0.347	[-1.182, +0.475]
RN50	D – D random	Relation	+4.916	[+3.927, +5.979]
RN50	D – D random	Both	-0.243	[-1.181, +0.678]

Table 17: Original retention contrasts, rows 65 through 80.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
ViT	Image main (0.0001)	I→T	+5.833	[+2.917, +8.800]
ViT	Reverse main (0.0001)	I→T	+0.167	[-0.650, +1.033]
ViT	Interaction (0.0001)	I→T	-0.067	[-1.667, +1.552]
ViT	Image main (0.0001)	T→I	+0.973	[+0.007, +1.930]
ViT	Reverse main (0.0001)	T→I	-1.093	[-2.040, -0.173]
ViT	Interaction (0.0001)	T→I	+0.400	[-0.560, +1.360]
ViT	Image main (0.0003)	I→T	+6.283	[+3.357, +9.333]
ViT	Reverse main (0.0003)	I→T	+0.850	[-0.500, +2.200]
ViT	Interaction (0.0003)	I→T	-0.367	[-2.767, +2.067]
ViT	Image main (0.0003)	T→I	+1.103	[-0.194, +2.382]
ViT	Reverse main (0.0003)	T→I	+0.223	[-0.825, +1.272]
ViT	Interaction (0.0003)	T→I	+0.047	[-1.247, +1.313]
ViT	Image main (0.001)	I→T	+9.517	[+6.407, +12.700]
ViT	Reverse main (0.001)	I→T	+1.317	[-0.450, +3.067]
ViT	Interaction (0.001)	I→T	+1.967	[-1.133, +5.067]
ViT	Image main (0.001)	T→I	+0.920	[-0.600, +2.417]

Table 18: Original directional contrasts at epoch ten, rows 1 through 16.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
ViT	Reverse main (0.001)	T→I	+0.993	[-0.342, +2.340]
ViT	Interaction (0.001)	T→I	+0.360	[-1.807, +2.477]
RN50	Image main (0.0001)	I→T	+4.917	[+1.717, +8.083]
RN50	Reverse main (0.0001)	I→T	+0.083	[-1.267, +1.433]
RN50	Interaction (0.0001)	I→T	-0.900	[-3.133, +1.352]
RN50	Image main (0.0001)	T→I	+0.150	[-1.030, +1.332]
RN50	Reverse main (0.0001)	T→I	+0.157	[-0.780, +1.080]
RN50	Interaction (0.0001)	T→I	+0.193	[-0.860, +1.264]
RN50	Image main (0.0003)	I→T	+5.683	[+2.483, +8.859]
RN50	Reverse main (0.0003)	I→T	+0.483	[-1.067, +2.017]
RN50	Interaction (0.0003)	I→T	-0.033	[-2.867, +2.800]
RN50	Image main (0.0003)	T→I	+0.177	[-1.205, +1.573]
RN50	Reverse main (0.0003)	T→I	+0.483	[-0.603, +1.590]
RN50	Interaction (0.0003)	T→I	-0.913	[-2.464, +0.647]
RN50	Image main (0.001)	I→T	+7.267	[+4.067, +10.567]
RN50	Reverse main (0.001)	I→T	+0.000	[-1.933, +1.967]

Table 19: Original directional contrasts at epoch ten, rows 17 through 32.

Encoder	Comparison	Endpoint	Difference	Adjusted interval
RN50	Interaction (0.001)	I→T	+1.800	$[-1.900, +5.433]$
RN50	Image main (0.001)	T→I	-0.360	$[-1.892, +1.163]$
RN50	Reverse main (0.001)	T→I	+1.533	$[+0.175, +2.870]$
RN50	Interaction (0.001)	T→I	-0.173	$[-2.173, +1.840]$

Table 20: Original directional contrasts at epoch ten, rows 33 through 36.

Encoder	Family	Tol.	e I→T	e T→I	Rel.	SC	Both	C I→T	C T→I
ViT	Source	1	84.50	68.96	81.04	83.02	66.76	55.71	39.20
ViT	U	1	84.50	69.76	79.47	81.21	68.95	56.32	39.35
ViT	A	1	83.83	69.03	84.22	83.47	67.43	54.75	39.15
ViT	D	1	83.10	70.21	84.99	82.91	69.31	54.17	39.54
ViT	AD	1	83.37	68.29	86.15	83.12	68.82	52.46	38.55
ViT	WiSE	1	83.80	68.87	84.56	83.35	69.27	53.85	38.98
ViT	AD random	1	83.23	68.10	81.87	82.54	67.72	53.84	38.45
ViT	Source	0	84.70	69.74	80.66	82.60	67.89	56.18	39.47
ViT	U	0	84.50	69.76	79.47	81.21	68.95	56.32	39.35
ViT	A	0	84.50	69.76	79.47	81.21	68.95	56.32	39.35
ViT	D	0	84.37	69.86	81.55	81.70	69.03	55.85	39.49
ViT	AD	0	84.47	69.49	86.16	83.22	68.97	53.91	39.02
ViT	WiSE	0	83.83	69.25	84.42	83.11	68.89	54.39	39.12
ViT	Frozen	–	84.50	69.76	79.47	81.21	68.95	56.32	39.35
RN50	Source	1	82.20	62.47	76.81	78.76	65.02	49.09	31.17
RN50	U	1	80.50	59.24	75.74	76.70	67.35	48.90	28.53
RN50	A	1	81.27	61.69	78.88	78.60	65.90	48.49	30.32
RN50	D	1	80.13	61.44	86.09	80.53	68.26	46.09	29.56

Table 21: AD selected procedures and descriptive zero-tolerance sensitivity. Entries are percentages. e denotes e-ViL; C denotes COCO. Rel. is relation accuracy; SC is SugarCrepe; Both is SugarCrepe++ both-caption accuracy.

Encoder	Family	Tol.	e I→T	e T→I	Rel.	SC	Both	C I→T	C T→I
RN50	AD	1	79.77	59.71	84.76	79.54	67.97	46.23	28.13
RN50	WiSE	1	80.30	63.36	84.92	80.82	67.37	47.09	31.26
RN50	AD random	1	81.21	59.99	80.49	79.21	67.86	47.58	28.36
RN50	Source	0	81.60	61.36	76.39	78.01	65.69	49.09	30.27
RN50	U	0	80.50	59.24	75.74	76.70	67.35	48.90	28.53
RN50	A	0	80.77	60.57	77.35	77.73	66.62	48.69	29.45
RN50	D	0	80.47	62.31	84.56	80.26	68.62	47.29	30.10
RN50	AD	0	82.27	60.60	84.77	80.03	68.46	48.01	28.81
RN50	WiSE	0	81.87	62.04	82.62	79.98	67.37	48.25	30.45
RN50	Frozen	–	80.50	59.24	75.74	76.70	67.35	48.90	28.53

Table 22: AD selected procedures and descriptive zero-tolerance sensitivity. Entries are percentages. e denotes e-ViL; C denotes COCO. Rel. is relation accuracy; SC is SugarCrepe; Both is SugarCrepe++ both-caption accuracy.

Encoder	Family	Seed	Draw	Rate	Epoch	λ	β	α	Norm
ViT	Source (1)	17	–	0.0001	2	1	0	1	0.9537
ViT	Source (1)	29	–	0.0001	3	1	0	1	1.331
ViT	Source (1)	43	–	0.0001	3	1	0	1	1.332
ViT	U (1)	17	–	0.0001	0	0	0	1	0
ViT	U (1)	29	–	0.0001	0	0	0	1	0
ViT	U (1)	43	–	0.0001	0	0	0	1	0
ViT	A (1)	17	–	0.0001	1	0.8	0	1	0.5879
ViT	A (1)	29	–	0.0001	3	0.8	0	1	1.366
ViT	A (1)	43	–	0.0001	4	0.8	0	1	1.716
ViT	D (1)	17	–	0.0001	4	0	4	1	1.156
ViT	D (1)	29	–	0.0001	5	0	4	1	1.325
ViT	D (1)	43	–	0.0001	3	0	4	1	0.9607
ViT	AD (1)	17	–	0.0003	7	0.5	1	1	3.242
ViT	AD (1)	29	–	0.0003	7	0.5	1	1	3.239
ViT	AD (1)	43	–	0.0003	9	0.5	1	1	3.649
ViT	WiSE (1)	17	–	0.0003	8	0	0	0.5	2.814
ViT	WiSE (1)	29	–	0.0003	10	0	0	0.5	3.154
ViT	WiSE (1)	43	–	0.0003	8	0	0	0.5	2.829
ViT	AD random (1)	17	0	0.0003	7	0.5	1	1	3.351
ViT	AD random (1)	17	1	0.0003	7	0.5	1	1	3.357

Table 23: AD study selected states. Family parentheses give the development tolerance in percentage points. Norm denotes the recorded parameter update norm. Dashes denote inapplicable recorded coefficients.

Encoder	Family	Seed	Draw	Rate	Epoch	λ	β	α	Norm
ViT	AD random (1)	17	2	0.0003	7	0.5	1	1	3.372
ViT	AD random (1)	29	0	0.0003	7	0.5	1	1	3.348
ViT	AD random (1)	29	1	0.0003	7	0.5	1	1	3.35
ViT	AD random (1)	29	2	0.0003	7	0.5	1	1	3.364
ViT	AD random (1)	43	0	0.0003	9	0.5	1	1	3.818
ViT	AD random (1)	43	1	0.0003	9	0.5	1	1	3.807
ViT	AD random (1)	43	2	0.0003	9	0.5	1	1	3.831
ViT	Source (0)	17	–	0.0001	1	1	0	1	0.571
ViT	Source (0)	29	–	0.0001	0	1	0	1	0
ViT	Source (0)	43	–	0.0001	2	1	0	1	0.9537
ViT	U (0)	17	–	0.0001	0	0	0	1	0
ViT	U (0)	29	–	0.0001	0	0	0	1	0
ViT	U (0)	43	–	0.0001	0	0	0	1	0
ViT	A (0)	17	–	0.0001	0	0.5	0	1	0
ViT	A (0)	29	–	0.0001	0	0.5	0	1	0
ViT	A (0)	43	–	0.0001	0	0.5	0	1	0
ViT	D (0)	17	–	0.0001	7	0	16	1	0.8569
ViT	D (0)	29	–	0.0001	1	0	16	1	0.4591
ViT	D (0)	43	–	0.0001	0	0	16	1	0
ViT	AD (0)	17	–	0.0001	10	0.5	1	1	2.522

Table 24: AD study selected states. Family parentheses give the development tolerance in percentage points. Norm denotes the recorded parameter update norm. Dashes denote inapplicable recorded coefficients.

Encoder	Family	Seed	Draw	Rate	Epoch	λ	β	α	Norm
ViT	AD (0)	29	–	0.0001	9	0.5	1	1	2.392
ViT	AD (0)	43	–	0.0001	9	0.5	1	1	2.381
ViT	WiSE (0)	17	–	0.0003	3	0	0	0.5	1.6
ViT	WiSE (0)	29	–	0.0003	5	0	0	0.5	2.15
ViT	WiSE (0)	43	–	0.0003	8	0	0	0.5	2.829
RN50	Source (1)	17	–	0.0001	2	1	0	1	1.732
RN50	Source (1)	29	–	0.0001	2	1	0	1	1.743
RN50	Source (1)	43	–	0.0001	2	1	0	1	1.719
RN50	U (1)	17	–	0.0001	0	0	0	1	0
RN50	U (1)	29	–	0.0001	0	0	0	1	0
RN50	U (1)	43	–	0.0001	0	0	0	1	0
RN50	A (1)	17	–	0.0001	0	0.8	0	1	0
RN50	A (1)	29	–	0.0001	2	0.8	0	1	1.763
RN50	A (1)	43	–	0.0001	2	0.8	0	1	1.734
RN50	D (1)	17	–	0.0001	7	0	1	1	3.193
RN50	D (1)	29	–	0.0001	5	0	1	1	2.564
RN50	D (1)	43	–	0.0001	6	0	1	1	2.932
RN50	AD (1)	17	–	0.0003	9	0.5	1	1	5.189
RN50	AD (1)	29	–	0.0003	6	0.5	1	1	4.217
RN50	AD (1)	43	–	0.0003	6	0.5	1	1	4.27

Table 25: AD study selected states. Family parentheses give the development tolerance in percentage points. Norm denotes the recorded parameter update norm. Dashes denote inapplicable recorded coefficients.

Encoder	Family	Seed	Draw	Rate	Epoch	λ	β	α	Norm
RN50	WiSE (1)	17	–	0.0001	5	0	0	0.75	2.572
RN50	WiSE (1)	29	–	0.0001	4	0	0	0.75	2.218
RN50	WiSE (1)	43	–	0.0001	4	0	0	0.75	2.216
RN50	AD random (1)	17	0	0.0003	9	0.5	1	1	5.389
RN50	AD random (1)	17	1	0.0003	9	0.5	1	1	5.376
RN50	AD random (1)	17	2	0.0003	9	0.5	1	1	5.35
RN50	AD random (1)	29	0	0.0003	6	0.5	1	1	4.315
RN50	AD random (1)	29	1	0.0003	6	0.5	1	1	4.309
RN50	AD random (1)	29	2	0.0003	6	0.5	1	1	4.305
RN50	AD random (1)	43	0	0.0003	6	0.5	1	1	4.356
RN50	AD random (1)	43	1	0.0003	6	0.5	1	1	4.341
RN50	AD random (1)	43	2	0.0003	6	0.5	1	1	4.342
RN50	Source (0)	17	–	0.0001	0	1	0	1	0
RN50	Source (0)	29	–	0.0001	2	1	0	1	1.743
RN50	Source (0)	43	–	0.0001	2	1	0	1	1.719
RN50	U (0)	17	–	0.0001	0	0	0	1	0
RN50	U (0)	29	–	0.0001	0	0	0	1	0
RN50	U (0)	43	–	0.0001	0	0	0	1	0
RN50	A (0)	17	–	0.0001	0	0.8	0	1	0
RN50	A (0)	29	–	0.0001	2	0.8	0	1	1.763

Table 26: AD study selected states. Family parentheses give the development tolerance in percentage points. Norm denotes the recorded parameter update norm. Dashes denote inapplicable recorded coefficients.

Encoder	Family	Seed	Draw	Rate	Epoch	λ	β	α	Norm
RN50	A (0)	43	–	0.0001	0	0.8	0	1	0
RN50	D (0)	17	–	0.0001	3	0	1	1	1.875
RN50	D (0)	29	–	0.0001	1	0	1	1	1.047
RN50	D (0)	43	–	0.0001	2	0	1	1	1.441
RN50	AD (0)	17	–	0.0001	9	0.5	1	1	3.09
RN50	AD (0)	29	–	0.0001	10	0.5	1	1	3.27
RN50	AD (0)	43	–	0.0001	10	0.5	1	1	3.272
RN50	WiSE (0)	17	–	0.0003	4	0	0	0.5	2.91
RN50	WiSE (0)	29	–	0.0003	4	0	0	0.5	2.901
RN50	WiSE (0)	43	–	0.0003	6	0	0	0.5	3.651

Table 27: AD study selected states. Family parentheses give the development tolerance in percentage points. Norm denotes the recorded parameter update norm. Dashes denote inapplicable recorded coefficients.

Encoder	Family	Tol.	e I→T	e T→I	Rel.	SC	Both	C I→T	C T→I
ViT	Source	1	84.50	68.96	81.04	83.02	66.76	55.71	39.20
ViT	U	1	84.50	69.76	79.47	81.21	68.95	56.32	39.35
ViT	A	1	83.83	69.03	84.22	83.47	67.43	54.75	39.15
ViT	D	1	83.10	70.21	84.99	82.91	69.31	54.17	39.54
ViT	WiSE	1	83.80	68.87	84.56	83.35	69.27	53.85	38.98
ViT	Image source	1	84.83	69.20	81.05	82.95	66.74	55.40	39.41
ViT	Reverse source	1	84.50	69.76	79.47	81.21	68.95	56.32	39.35
ViT	D random	1	83.33	70.41	82.14	82.45	68.70	54.96	39.52
ViT	Source	0	84.70	69.74	80.66	82.60	67.89	56.18	39.47
ViT	U	0	84.50	69.76	79.47	81.21	68.95	56.32	39.35
ViT	A	0	84.50	69.76	79.47	81.21	68.95	56.32	39.35
ViT	D	0	84.37	69.86	81.55	81.70	69.03	55.85	39.49
ViT	WiSE	0	83.83	69.25	84.42	83.11	68.89	54.39	39.12
ViT	Image source	0	84.83	69.47	81.03	83.00	66.96	55.65	39.49
ViT	Reverse source	0	84.50	69.76	79.47	81.21	68.95	56.32	39.35
ViT	Frozen	–	84.50	69.76	79.47	81.21	68.95	56.32	39.35
RN50	Source	1	82.20	62.47	76.81	78.76	65.02	49.09	31.17
RN50	U	1	80.50	59.24	75.74	76.70	67.35	48.90	28.53

Table 28: Original selected procedures and descriptive zero-tolerance sensitivity. Entries are percentages. e denotes e-ViL; C denotes COCO. Rel. is relation accuracy; SC is SugarCrepe; Both is SugarCrepe++ both-caption accuracy.

Encoder	Family	Tol.	e I→T	e T→I	Rel.	SC	Both	C I→T	C T→I
RN50	A	1	81.27	61.69	78.88	78.60	65.90	48.49	30.32
RN50	D	1	80.13	61.44	86.09	80.53	68.26	46.09	29.56
RN50	WiSE	1	80.30	63.36	84.92	80.82	67.37	47.09	31.26
RN50	Image source	1	82.40	62.85	77.09	78.70	65.03	49.20	31.49
RN50	Reverse source	1	80.50	59.24	75.74	76.70	67.35	48.90	28.53
RN50	D random	1	80.87	61.79	81.18	80.21	68.50	47.63	29.88
RN50	Source	0	81.60	61.36	76.39	78.01	65.69	49.09	30.27
RN50	U	0	80.50	59.24	75.74	76.70	67.35	48.90	28.53
RN50	A	0	80.77	60.57	77.35	77.73	66.62	48.69	29.45
RN50	D	0	80.47	62.31	84.56	80.26	68.62	47.29	30.10
RN50	WiSE	0	81.87	62.04	82.62	79.98	67.37	48.25	30.45
RN50	Image source	0	81.83	61.65	76.56	77.95	65.60	49.13	30.52
RN50	Reverse source	0	80.50	59.24	75.74	76.70	67.35	48.90	28.53
RN50	Frozen	–	80.50	59.24	75.74	76.70	67.35	48.90	28.53

Table 29: Original selected procedures and descriptive zero-tolerance sensitivity. Entries are percentages. e denotes e-ViL; C denotes COCO. Rel. is relation accuracy; SC is SugarCrepe; Both is SugarCrepe++ both-caption accuracy.

Encoder	Family	Seed	Draw	Rate	Epoch	λ	β	α	Norm
ViT	Source (1)	17	–	0.0001	2	–	–	1	0.9537
ViT	Source (1)	29	–	0.0001	3	–	–	1	1.331
ViT	Source (1)	43	–	0.0001	3	–	–	1	1.332
ViT	U (1)	17	–	0.0001	0	–	–	1	0
ViT	U (1)	29	–	0.0001	0	–	–	1	0
ViT	U (1)	43	–	0.0001	0	–	–	1	0
ViT	A (1)	17	–	0.0001	1	0.8	–	1	0.5879
ViT	A (1)	29	–	0.0001	3	0.8	–	1	1.366
ViT	A (1)	43	–	0.0001	4	0.8	–	1	1.716
ViT	D (1)	17	–	0.0001	4	–	4	1	1.156
ViT	D (1)	29	–	0.0001	5	–	4	1	1.325
ViT	D (1)	43	–	0.0001	3	–	4	1	0.9607
ViT	WiSE (1)	17	–	0.0003	8	–	–	0.5	2.814
ViT	WiSE (1)	29	–	0.0003	10	–	–	0.5	3.154
ViT	WiSE (1)	43	–	0.0003	8	–	–	0.5	2.829
ViT	Image source (1)	17	–	0.0001	2	–	–	1	0.9775
ViT	Image source (1)	29	–	0.0001	3	–	–	1	1.349
ViT	Image source (1)	43	–	0.0001	3	–	–	1	1.346
ViT	Reverse source (1)	17	–	0.0001	0	–	–	1	0
ViT	Reverse source (1)	29	–	0.0001	0	–	–	1	0

Table 30: Original study selected states. Family parentheses give the development tolerance in percentage points. Norm denotes the recorded parameter update norm. Dashes denote inapplicable recorded coefficients.

Encoder	Family	Seed	Draw	Rate	Epoch	λ	β	α	Norm
ViT	Reverse source (1)	43	–	0.0001	0	–	–	1	0
ViT	D random (1)	17	0	0.0001	4	–	4	1	1.161
ViT	D random (1)	17	1	0.0001	4	–	4	1	1.169
ViT	D random (1)	17	2	0.0001	4	–	4	1	1.169
ViT	D random (1)	29	0	0.0001	5	–	4	1	1.341
ViT	D random (1)	29	1	0.0001	5	–	4	1	1.339
ViT	D random (1)	29	2	0.0001	5	–	4	1	1.341
ViT	D random (1)	43	0	0.0001	3	–	4	1	0.9626
ViT	D random (1)	43	1	0.0001	3	–	4	1	0.9629
ViT	D random (1)	43	2	0.0001	3	–	4	1	0.963
ViT	Source (0)	17	–	0.0001	1	–	–	1	0.571
ViT	Source (0)	29	–	0.0001	0	–	–	1	0
ViT	Source (0)	43	–	0.0001	2	–	–	1	0.9537
ViT	U (0)	17	–	0.0001	0	–	–	1	0
ViT	U (0)	29	–	0.0001	0	–	–	1	0
ViT	U (0)	43	–	0.0001	0	–	–	1	0
ViT	A (0)	17	–	0.0001	0	0.5	–	1	0
ViT	A (0)	29	–	0.0001	0	0.5	–	1	0
ViT	A (0)	43	–	0.0001	0	0.5	–	1	0
ViT	D (0)	17	–	0.0001	7	–	16	1	0.8569

Table 31: Original study selected states. Family parentheses give the development tolerance in percentage points. Norm denotes the recorded parameter update norm. Dashes denote inapplicable recorded coefficients.

Encoder	Family	Seed	Draw	Rate	Epoch	λ	β	α	Norm
ViT	D (0)	29	–	0.0001	1	–	16	1	0.4591
ViT	D (0)	43	–	0.0001	0	–	16	1	0
ViT	WiSE (0)	17	–	0.0003	3	–	–	0.5	1.6
ViT	WiSE (0)	29	–	0.0003	5	–	–	0.5	2.15
ViT	WiSE (0)	43	–	0.0003	8	–	–	0.5	2.829
ViT	Image source (0)	17	–	0.0001	1	–	–	1	0.5897
ViT	Image source (0)	29	–	0.0001	3	–	–	1	1.349
ViT	Image source (0)	43	–	0.0001	3	–	–	1	1.346
ViT	Reverse source (0)	17	–	0.0001	0	–	–	1	0
ViT	Reverse source (0)	29	–	0.0001	0	–	–	1	0
ViT	Reverse source (0)	43	–	0.0001	0	–	–	1	0
RN50	Source (1)	17	–	0.0001	2	–	–	1	1.732
RN50	Source (1)	29	–	0.0001	2	–	–	1	1.743
RN50	Source (1)	43	–	0.0001	2	–	–	1	1.719
RN50	U (1)	17	–	0.0001	0	–	–	1	0
RN50	U (1)	29	–	0.0001	0	–	–	1	0
RN50	U (1)	43	–	0.0001	0	–	–	1	0
RN50	A (1)	17	–	0.0001	0	0.8	–	1	0
RN50	A (1)	29	–	0.0001	2	0.8	–	1	1.763
RN50	A (1)	43	–	0.0001	2	0.8	–	1	1.734

Table 32: Original study selected states. Family parentheses give the development tolerance in percentage points. Norm denotes the recorded parameter update norm. Dashes denote inapplicable recorded coefficients.

Encoder	Family	Seed	Draw	Rate	Epoch	λ	β	α	Norm
RN50	D (1)	17	–	0.0001	7	–	1	1	3.193
RN50	D (1)	29	–	0.0001	5	–	1	1	2.564
RN50	D (1)	43	–	0.0001	6	–	1	1	2.932
RN50	WiSE (1)	17	–	0.0001	5	–	–	0.75	2.572
RN50	WiSE (1)	29	–	0.0001	4	–	–	0.75	2.218
RN50	WiSE (1)	43	–	0.0001	4	–	–	0.75	2.216
RN50	Image source (1)	17	–	0.0001	2	–	–	1	1.744
RN50	Image source (1)	29	–	0.0001	2	–	–	1	1.742
RN50	Image source (1)	43	–	0.0001	2	–	–	1	1.747
RN50	Reverse source (1)	17	–	0.0001	0	–	–	1	0
RN50	Reverse source (1)	29	–	0.0001	0	–	–	1	0
RN50	Reverse source (1)	43	–	0.0001	0	–	–	1	0
RN50	D random (1)	17	0	0.0001	7	–	1	1	3.379
RN50	D random (1)	17	1	0.0001	7	–	1	1	3.375
RN50	D random (1)	17	2	0.0001	7	–	1	1	3.356
RN50	D random (1)	29	0	0.0001	5	–	1	1	2.68
RN50	D random (1)	29	1	0.0001	5	–	1	1	2.661
RN50	D random (1)	29	2	0.0001	5	–	1	1	2.652
RN50	D random (1)	43	0	0.0001	6	–	1	1	3.075
RN50	D random (1)	43	1	0.0001	6	–	1	1	3.069

Table 33: Original study selected states. Family parentheses give the development tolerance in percentage points. Norm denotes the recorded parameter update norm. Dashes denote inapplicable recorded coefficients.

Encoder	Family	Seed	Draw	Rate	Epoch	λ	β	α	Norm
RN50	D random (1)	43	2	0.0001	6	–	1	1	3.06
RN50	Source (0)	17	–	0.0001	0	–	–	1	0
RN50	Source (0)	29	–	0.0001	2	–	–	1	1.743
RN50	Source (0)	43	–	0.0001	2	–	–	1	1.719
RN50	U (0)	17	–	0.0001	0	–	–	1	0
RN50	U (0)	29	–	0.0001	0	–	–	1	0
RN50	U (0)	43	–	0.0001	0	–	–	1	0
RN50	A (0)	17	–	0.0001	0	0.8	–	1	0
RN50	A (0)	29	–	0.0001	2	0.8	–	1	1.763
RN50	A (0)	43	–	0.0001	0	0.8	–	1	0
RN50	D (0)	17	–	0.0001	3	–	1	1	1.875
RN50	D (0)	29	–	0.0001	1	–	1	1	1.047
RN50	D (0)	43	–	0.0001	2	–	1	1	1.441
RN50	WiSE (0)	17	–	0.0003	4	–	–	0.5	2.91
RN50	WiSE (0)	29	–	0.0003	4	–	–	0.5	2.901
RN50	WiSE (0)	43	–	0.0003	6	–	–	0.5	3.651
RN50	Image source (0)	17	–	0.0001	0	–	–	1	0
RN50	Image source (0)	29	–	0.0001	2	–	–	1	1.742
RN50	Image source (0)	43	–	0.0001	2	–	–	1	1.747
RN50	Reverse source (0)	17	–	0.0001	0	–	–	1	0

Table 34: Original study selected states. Family parentheses give the development tolerance in percentage points. Norm denotes the recorded parameter update norm. Dashes denote inapplicable recorded coefficients.

Encoder	Family	Seed	Draw	Rate	Epoch	λ	β	α	Norm
RN50	Reverse source (0)	29	–	0.0001	0	–	–	1	0
RN50	Reverse source (0)	43	–	0.0001	0	–	–	1	0

Table 35: Original study selected states. Family parentheses give the development tolerance in percentage points. Norm denotes the recorded parameter update norm. Dashes denote inapplicable recorded coefficients.

C Full-Gallery Certificate Diagnostics

These diagnostics use the original retention selections without changing their benchmark ranks or practical criteria. Each distinct selected state appears once, even when several selection rules choose it. Different selected states reuse the same queries. The pooled denominator therefore counts state-query observations, not independent test examples. The complete artifact retains each selected state's selection roles and all query-level certificate masks. These descriptive counts do not add statistical hypotheses or imply an unseen-gallery guarantee. Computed KL values and thresholds use floating-point arithmetic. The audit checks numerical agreement and observed rank consistency, rather than interval-arithmetic guarantees near boundaries.

Encoder	Pool	Direction	Group	States	Teacher	Retained	KL	Score	Union
RN50	e-ViL	I→T	Trained	20	16,100	15,021	7.09	4.30	7.99
RN50	e-ViL	I→T	Unchanged	10	8,050	8,050	100.00	100.00	100.00
RN50	e-ViL	T→I	Trained	20	59,240	55,185	35.57	11.43	35.60
RN50	e-ViL	T→I	Unchanged	10	29,620	29,620	100.00	100.00	100.00
RN50	COCO	I→T	Trained	20	48,900	41,425	2.85	2.64	3.74
RN50	COCO	I→T	Unchanged	10	24,450	24,450	100.00	100.00	100.00
RN50	COCO	T→I	Trained	20	142,640	127,374	18.43	5.69	18.51
RN50	COCO	T→I	Unchanged	10	71,320	71,320	100.00	100.00	100.00
ViT	e-ViL	I→T	Trained	22	18,590	17,834	45.11	13.17	45.11
ViT	e-ViL	I→T	Unchanged	11	9,295	9,295	100.00	100.00	100.00
ViT	e-ViL	T→I	Trained	22	76,736	72,944	67.72	25.35	67.72
ViT	e-ViL	T→I	Unchanged	11	38,368	38,368	100.00	100.00	100.00
ViT	COCO	I→T	Trained	22	61,952	55,980	29.67	7.43	29.67
ViT	COCO	I→T	Unchanged	11	30,976	30,976	100.00	100.00	100.00
ViT	COCO	T→I	Trained	22	216,414	200,619	55.31	18.13	55.31
ViT	COCO	T→I	Unchanged	11	108,207	108,207	100.00	100.00	100.00

Table 36: Full-gallery certificate diagnostics for all primary and sensitivity selections in the original retention study. Each selected state counts once. Trained requires positive epoch and update norm; other states appear as Unchanged. Teacher and Retained count state-query observations, not distinct queries. KL, Score, and Union give certified percentages of teacher-correct observations under pessimistic ties.

D Exact Gallery-Level Rank-Retention Geometry

This section gives an exact distance to failure for a fixed retrieval problem. It also gives a complementary score bound and an aggregate KL budget bound. None changes the empirical selection rule or primary R@1 endpoints. Surrogate calibration already connects surrogate losses with decision errors (Bartlett et al., 2003; Ávila Pires and Szepesvári, 2016). Here the specific result is the exact positive-set failure radius for a fixed retrieval gallery. The proof uses standard convex projection arguments.

D.1 Definitions

Fix one query and a finite gallery with n candidates. Let R be its nonempty relevant set and N its irrelevant set. Teacher and student distributions q, p lie in the same probability simplex. Define forward KL in nats by

$$D(q||p) = \sum_{i=1}^n q_i \log \frac{q_i}{p_i}. \quad (17)$$

Use $0 \log(0/p) = 0$. A positive numerator with zero denominator contributes infinity. For softmax probabilities, both models use the same gallery and positive temperature.

For an integer $K \geq 1$, define the closed failure event

$$F_K = \left\{ p : \left| \{j \in N : p_j \geq \max_{i \in R} p_i\} \right| \geq K \right\}. \quad (18)$$

Cross-relevance ties count against the relevant candidate. Outside F_K , every tie rule places a relevant item within the first K positions. If $|N| < K$, failure is impossible.

Write $a_1 \geq \dots \geq a_r$ for relevant teacher probabilities. Write $b_1 \geq \dots \geq b_s$ for irrelevant teacher probabilities. When $K \leq s$, strict teacher correctness means $a_1 > b_K$. The exact distance to failure is

$$C_K(q, R) = \min_{p \in F_K} D(q||p). \quad (19)$$

D.2 Exact positive-set projection

Theorem 1. *If $|N| < K$, then $C_K = +\infty$. If $a_1 \leq b_K$, then $C_K = 0$. Otherwise, let S contain the K largest irrelevant teacher probabilities. The unique threshold t solves*

$$\sum_{i \in R} (q_i - t)_+ = \sum_{j \in S} (t - q_j)_+. \quad (20)$$

Define $A = \{i \in R : q_i > t\}$ and $B = \{j \in S : q_j < t\}$. An attaining distribution is

$$p_i^* = \begin{cases} \min(q_i, t), & i \in R, \\ \max(q_i, t), & i \in S, \\ q_i, & i \notin R \cup S. \end{cases} \quad (21)$$

Its pooling level and divergence are

$$t = \frac{\sum_{i \in A} q_i + \sum_{j \in B} q_j}{|A| + |B|}, \quad (22)$$

$$C_K = \sum_{i \in A} q_i \log \frac{q_i}{t} + \sum_{j \in B} q_j \log \frac{q_j}{t}. \quad (23)$$

Coordinates equal to t can join either active set without changing the result. Zero-probability selected candidates can receive positive mass and contribute zero directly to KL.

Proof. For a fixed K -element set S , failure requires $p_j \geq p_i$ for all $j \in S$ and $i \in R$. The complete event is the union over such sets.

First consider which irrelevant candidates to select. Suppose $u \in S$, $v \in N \setminus S$, and $q_v \geq q_u$. Take any feasible finite-cost distribution p . If $p_v \geq p_u$, replacing u by v preserves feasibility without changing p . Otherwise, swap p_u and p_v . The swapped distribution is feasible for the replacement set. Its cost change is

$$(q_v - q_u) \log(p_v/p_u) \leq 0. \quad (24)$$

The boundary convention extends this argument to zero coordinates. Repeated exchanges select the K largest irrelevant teacher probabilities without increasing the minimum cost.

For this fixed set, introduce t with constraints $p_i \leq t$ for relevant coordinates. Selected irrelevant coordinates satisfy $p_j \geq t$. Up to a constant, the objective is the convex function $-\sum_i q_i \log p_i$. The proposed projection preserves total mass by Equation 20. It also satisfies every order constraint.

Use simplex multiplier one. For an active relevant coordinate, use multiplier $\mu_i = q_i/t - 1 \geq 0$ on $p_i - t \leq 0$. For an active selected irrelevant coordinate, use $\nu_j = 1 - q_j/t \geq 0$ on $t - p_j \leq 0$. All inactive order multipliers vanish. Stationarity in each positive coordinate follows directly. Stationarity in t requires

$$\sum_{i \in A} \mu_i = \sum_{j \in B} \nu_j. \quad (25)$$

This equality is the pooled-mean identity. For an unpooled zero coordinate, nonnegativity multiplier

one satisfies boundary stationarity. Complementarity holds. These convex optimality conditions prove minimality.

The balance function is continuous. It is positive at the smallest selected irrelevant value and negative at the largest relevant value. Between those bounds, at least one active coordinate makes it strictly decreasing. Thus its root exists and is unique. Substituting the projection into KL gives Equation 23. $\square$

Corollary 1. *If $D(q\|p) < C_K(q, R)$, the student retains a relevant item within the first K positions. The open radius is optimal.*

Proof. The inequality excludes the failure set by the definition of C_K . The attaining projection reaches failure at equality under pessimistic ties. Strict failures can approach that projection arbitrarily closely. Thus no larger open KL radius works in general. $\square$

D.3 R@1 and boundary behavior

For $K = 1$, let b be the largest irrelevant teacher probability. Start a pool with b . Add relevant probabilities in decreasing order while the next value exceeds the current mean. If m relevant coordinates join, then

$$t = \frac{b + \sum_{i=1}^m a_i}{m + 1}, \quad (26)$$

$$C_1 = \sum_{i=1}^m a_i \log(a_i/t) + b \log(b/t). \quad (27)$$

Adding a relevant value raises the mean without exceeding that value. Previously active coordinates therefore remain active. This proves the one-pass pooling procedure.

The R@1 cost is $O(n + r \log r)$. General K first selects the top K irrelevant values. The balance function is linear between its probability breakpoints. The implementation scans active sets with cost $O((r + K)^2)$ after the gallery scan. Relevant sets contain five captions or one image in the source retrieval tasks.

For $q = (.35, .30, .25, .10)$ and $R = \{1, 2, 3\}$, the exact radius is approximately .08083268 nats. The two-coordinate radius is approximately .07354844 nats. This deterministic example illustrates the contribution of multiple relevant candidates. It is not an empirical retrieval result.

Whenever failure is possible, $C_K \leq \log(K + 1)$. To see this, select any K irrelevant indices and

define

$$p = \frac{q + \sum_{j \in S} e_j}{K + 1}. \quad (28)$$

Each selected irrelevant probability is at least every relevant probability. Also $p_i \geq q_i/(K + 1)$, which bounds the divergence. A point mass on one relevant candidate attains the bound.

D.4 An exact score radius

Let z and $z + d$ be teacher and student scores on the same gallery. Let $z_{(K,N)}$ be the K th largest irrelevant score. Define

$$g_K = \max_{i \in R} z_i - z_{(K,N)}, \quad (29)$$

$$\begin{aligned} \text{osc}(d) &= \max_i d_i - \min_i d_i \\ &= 2 \min_c \|d - c\mathbf{1}\|_\infty. \end{aligned} \quad (30)$$

Theorem 2. *For a strictly correct teacher, $\text{osc}(d) < g_K$ preserves a relevant item within the first K positions. The exact failure radius under the centered infinity norm is $g_K/2$.*

Proof. Choose a teacher-best relevant candidate i^* . Every irrelevant j with $z_j \leq z_{(K,N)}$ satisfies

$$(z_{i^*} + d_{i^*}) - (z_j + d_j) \geq g_K - \text{osc}(d) > 0. \quad (31)$$

At most $K - 1$ irrelevant candidates lie above that threshold. This proves retention.

For sharpness, decrease every relevant score by $g_K/2$. Increase the K largest irrelevant scores by $g_K/2$. Leave all other scores unchanged. The selected irrelevant candidates now tie or exceed every relevant candidate. The oscillation is exactly g_K . $\square$

This radius removes harmless common score shifts. It improves the uncentered sufficient condition $2\|d\|_\infty < g_K$. Consistent temperature scaling preserves the comparison.

For $q = \text{softmax}(z/T)$ and $p = \text{softmax}((z + d)/T)$,

$$D(q\|p) = \log \mathbb{E}_q e^{d/T} - \mathbb{E}_q [d/T]. \quad (32)$$

Hoeffding's bounded-variable inequality gives

$$D(q\|p) \leq \frac{\text{osc}(d)^2}{8T^2}. \quad (33)$$

Neither sufficient certificate dominates the other on every query. Their union remains a valid certificate.

D.5 A total KL budget

Consider teacher-correct queries with positive finite thresholds C_i . Suppose their full-gallery divergences sum to at most B . Sort thresholds increasingly as $C_{(1)} \leq \dots \leq C_{(m)}$. The number of lost queries cannot exceed

$$\max \left\{ h : \sum_{i=1}^h C_{(i)} \leq B \right\}. \quad (34)$$

Each lost query costs at least its failure threshold. Buying the cheapest thresholds therefore gives the largest possible failure count. The bound is sharp when query distributions can vary independently. Theorem 1 supplies an attaining failure distribution for every threshold. A shared embedding model can impose additional constraints. The bound remains valid there, but need not be attainable.

D.6 Scope and numerical safeguards

The certificate uses the actual query, candidate gallery, relevance set, and temperature. It cannot replace any of these with a training surrogate. Two counterexamples make the boundary explicit.

First, take teacher scores $(2, 1, 0)$ with the first candidate relevant. Train only on candidates one and two. Student scores $(2, 1, 10)$ give zero KL on that training subgallery. The full gallery nevertheless retrieves the third, irrelevant candidate.

Second, give 100 queries teacher probabilities $(.51, .49)$, with the first candidate relevant. Leave 99 unchanged and switch one student to $(.49, .51)$. Average KL can fall below the individual threshold while one query fails. An average divergence cannot replace each query’s divergence in the individual certificate.

The implementation rejects unsafe overflow rather than returning a passing certificate. It uses the conservative comparison

$$D + 10^{-12} + 10^{-10}|C_K| < C_K. \quad (35)$$

These tolerances do not constitute interval arithmetic. The reported quantities remain floating-point diagnostics. Benchmark ranking can use a fixed index tie rule. Certificate correctness uses pessimistic ties and must remain separately labeled.

Any empirical coverage report must state its denominator. Useful denominators include all queries, strictly correct teacher queries, and actually retained teacher-correct queries. Frozen states must remain visible when interpreting coverage.

The present proofs do not assert a measured coverage gain.

Rate	Positives	Relation	Edited	Both
–	Frozen	79.47	81.21	68.95
10^{-4}	Source	77.75	81.00	64.21
10^{-4}	Supported	87.62	83.93	67.75
$3 \cdot 10^{-4}$	Source	75.35	80.14	63.77
$3 \cdot 10^{-4}$	Supported	86.68	83.35	67.16
10^{-3}	Source	70.51	76.35	60.25
10^{-3}	Supported	85.78	81.28	65.62

Table 37: Mean accuracy (%) at matched epoch ten. Relation is e-SNLI-VE; Edited is SugarCrep; Both requires both valid SugarCrep++ captions above the foil. These are descriptive endpoints.

E Earlier Matched Assignment Study: Overview

The following appendices preserve the completed 72-fit study and its original inferential family. Its comparisons use the ViT linear adapter and precede the replication and AD extension. Statements about unexecuted allocation refer to that earlier protocol. The new extension uses the separate methods in Appendix A. The original tables and figures remain unchanged. Its manuscript, aggregates, and historical audit receipts survive recovery. Its original raw prediction archives are unavailable in the current workspace. Current strengthening results use separate, newly audited artifacts.

F Matched Assignment Study: Complete Methods

F.1 Protocol and comparison

The matched assignment study follows the earlier twelve-policy experiment. Its scientific choices were frozen before its own fitting and scoring in docs/REVIEW_FOLLOWUP_PROTOCOL.json. It asks what label-informed positive assignment contributes beyond positive count, coarse

Selection	Positives	Epochs	I→T	T→I
Native	Source	1/1/3	84.70	69.35
Native	Supported	5/10/5	77.00	63.30
Retrieval	Source	1/2/2	84.83	69.70
Retrieval	Supported	0/0/0	84.50	69.76

Table 38: Selected procedures on the full source-caption test pool. Epochs follow seeds 17/29/43. Native supported uses learning rate $3 \cdot 10^{-4}$; the other rows use 10^{-4} . Epoch zero is frozen initialization.

frozen-score difficulty, and training schedule. The earlier protocol, outcomes, and inferential family remain separate. All four policy families use the same source captions and hypotheses, unit candidate weights, uniform positive targets, and the original symmetric mean-log-probability loss. The source policy makes only original captions positive. Supported promotion adds hypotheses labeled supported. The two randomized controls replace this added set, as defined below. Hypotheses that are not promoted remain candidates; they are not removed from the denominator. Original labels and evaluation targets are never overwritten.

F.2 Fixed promotion assignments

For training image i , let Q_i contain all its annotated hypotheses, S_i its source captions, and $H_i \subseteq Q_i$ its supported hypotheses. Write $K_i = |S_i|$ and $M_i = |H_i|$. The count-only control samples an M_i -element subset of Q_i uniformly without replacement. The score-stratified control first computes the cosine between each hypothesis and its owning image from cached frozen features, using explicit L2 normalization and float64 arithmetic. It orders hypotheses by increasing cosine, breaking exact ties by global manifest text index, and splits the ordered list into two bins with `numpy.array_split`; the lower-score bin receives the extra element for an odd count. In each bin, it uniformly samples exactly the number of hypotheses labeled supported originally in that bin. Thus the strata do not use labels, but their quotas do.

Each control has three independent assignment draws, seeded 101, 211, and 307 using NumPy PCG64. Images are processed in manifest order. Each complete draw is fixed across all training seeds, learning rates, batches, and epochs. Empty and fully selected groups are retained. Naturally unchanged assignments are not redrawn. Distinct text IDs remain distinct candidates even when their strings coincide. Assignment records retain original labels, sampled IDs, label overlap, identity-level and exact-string changes, bin quotas, forced versus exchangeable groups, exact score ties, and frozen-score imbalance. Two bins match coarse score strata, not exact caption scores. Because their counts are label-derived, these controls are identification devices rather than annotation-free alternatives.

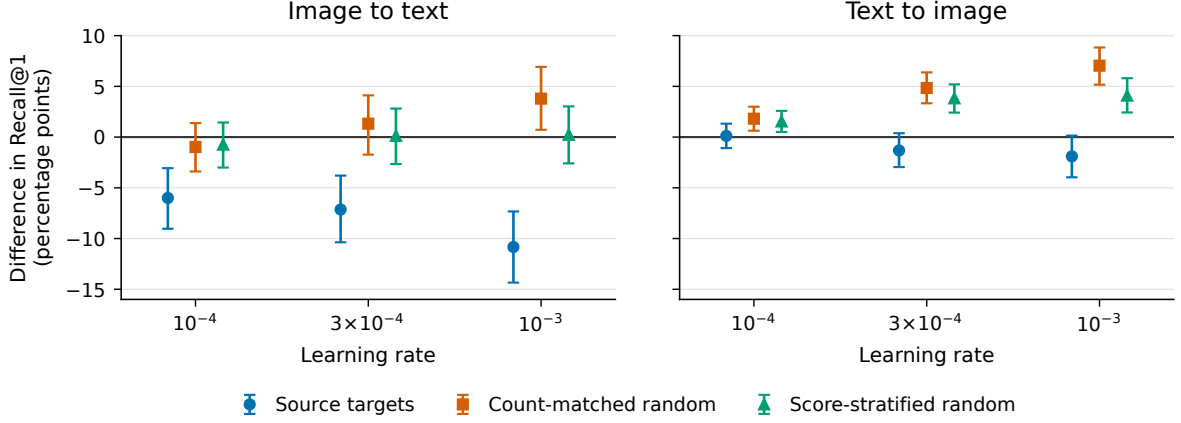


Figure 2: All 18 primary comparisons at matched epoch ten. Points are supported promotion minus the named comparator. Bars are 99.7222% paired image-cluster intervals, adjusted for the 18 comparisons and conditional on the fitted seeds and assignment draws. Random controls average three fixed assignments within each of three training seeds.

F.3 What the matched controls preserve

For any paired batch, all three expanded policies retain the same images, candidate IDs and strings, source positives, unit denominator weights, and $K_i + M_i$ positives per image. Each annotated text ID has exactly one owning image. Hence they also have the same number of eligible reverse anchors,

$$T_+ = \sum_{i=1}^B (K_i + M_i), \quad (36)$$

where B is the number of batch images. The identities of promoted hypothesis anchors can differ; their total count cannot. This holds for every batch, including the final partial batch.

A stronger fixed-score invariant follows. Let z_{ij} be the scaled image-text score, $p_{ij} = e^{z_{ij}} / \sum_k e^{z_{ik}}$ the image-row softmax, and $r_{ij} = e^{z_{ij}} / \sum_h e^{z_{hj}}$ the text-column softmax. Both sums use the unchanged complete candidate pools. Let P_i be the chosen positive captions for image i , E the eligible caption anchors, and $o(j)$ the owning image of eligible caption j . The full symmetric loss has derivative

$$\begin{aligned} \frac{\partial L}{\partial z_{ij}} = & \frac{1}{2B} \left(p_{ij} - \frac{\mathbf{1}\{j \in P_i\}}{K_i + M_i} \right) \\ & + \frac{\mathbf{1}\{j \in E\}}{2T_+} (r_{ij} - \mathbf{1}\{i = o(j)\}), \end{aligned} \quad (37)$$

where the second term is defined as zero for $j \notin E$. For any source-caption column, every quantity on the right is identical across supported, count-only, and score-stratified promotion at common

logits. Their entire source-column score gradients therefore agree. The proof uses the unchanged source indicators, equal row-positive counts, equal reverse-anchor counts, and unchanged softmax distributions. Gradients on other columns can differ. Shared adapter parameters combine all columns, so this identity does not imply identical parameter updates or source scores after training. It also does not compare expanded policies with source-only training, whose positive and reverse-anchor counts differ.

Expansion changes both directions. Within an image row, source targets change from $1/K_i$ to $1/(K_i + M_i)$. The reverse direction changes differently. Let T_0 be the number of original source-caption anchors, H the disjoint set of newly eligible hypothesis anchors, and ℓ_j^T a caption's loss against the unchanged image pool. Each old caption retains its positive-image target and its individual loss at fixed scores, but

$$L_T^{\text{expanded}} = \frac{T_0 L_T^{\text{source}} + \sum_{j \in H} \ell_j^T}{T_0 + |H|}. \quad (38)$$

Thus original anchors retain fraction $T_0/(T_0 + |H|)$ of the reverse average. This is not an undiluted reverse objective. The fixed-score identities describe distinct task structures; they do not establish which change causes a trained model's directional retrieval outcome.

F.4 Crossed fitting grid and replay

The original 1,200 training images and all their source captions and hypotheses are unchanged.

Two deterministic policies and six randomized policy draws form eight fitting conditions. Crossing these with learning rates 10^{-4} , 3×10^{-4} , 10^{-3} and training seeds 17, 29, and 43 yields 72 fits. Assignment draws and training seeds are separate factors. Every condition uses the same image order for a given training seed, with batches of 32 images and ten epochs. Epochs 0 through 10 are retained, giving 792 state references. Epoch 10 defines the primary matched-schedule comparison; epochs 1 and 5 provide descriptive trajectories.

The frozen encoder, pretrained score scale, and zero-initialized independent 512×512 residual maps are inherited from the original study. AdamW uses weight decay .01, coefficients (.9, .999), epsilon 10^{-8} , gradient-norm clipping at one, and no scheduler; AMSGrad, fused, and foreach updates are disabled. Fitting uses deterministic CPU operations with two threads. Source and supported fitting call the unchanged original loss implementations. The randomized masks call that same multipositive implementation after changing only training-positive assignments. All 18 source/supported candidates must reproduce the original training losses and native validation histories exactly, together with exact tensors at their original best epochs; a discrepancy stops execution. Immutable ledgers bind the protocol, original sources, features, six assignment files, and new code before fitting. Completion receipts bind every retained epoch state and validation artifact.

F.5 Source-caption retrieval pools

Source-only development retrieval uses the official e-ViL development image pool after excluding the original 100 calibration images: 900 images and 4,500 source captions. It includes the original 100 native-validation images. The test pool contains all 1,000 official e-ViL test images and their 5,000 source captions. Each image has five relevant source captions, and each caption has one relevant owning image. Hypotheses are absent from these evaluation candidate pools. Both directions search their complete pools. These are e-ViL-pool Flickr source-caption measurements, not the standard Karpathy Flickr1k split, which overlaps the original fitting and validation data.

Image IDs and image-file SHA256 values are checked for forbidden overlap across training, calibration, source development, and source test. The intentional inclusion of native validation within the development pool and of the original 400 relation-

test images within the 1,000-image test pool is recorded. Source-caption provenance, repeated strings, and string overlap are audited separately. Cached features are reused only under matching IDs or exact content and encoder provenance. The original 400 test images and additional 600 images are reported as descriptive query subgroups using the same full candidate pool, not as separately selected benchmarks. The original VE400, Sugar-Crepe, SugarCrepe++, and COCO pools remain unchanged.

F.6 Two development selectors

The native selector exactly reuses the earlier criterion: the mean of source-or-supported image-to-text R@1 and image-balanced supported-versus-contradicted accuracy on the original 100 validation images and 1,875 caption IDs. Relation ties receive half credit. The source-retrieval selector averages image-to-text and text-to-image R@1 on the source-only development pool above. For each policy or assignment draw and each learning rate, it chooses the best epoch separately for each training seed, including epoch 0; exact ties retain the earliest epoch. It then chooses one learning rate by the mean best score across the three training seeds, breaking ties toward the lower rate.

Each randomized draw is selected separately and the three draws receive equal weight in reporting. No best draw is selected. The two selectors define complete selection procedures: they differ in criterion, relevance, candidate pool, direction balance, and development-set size. Their comparison does not isolate the criterion alone. Selection references existing epoch states instead of duplicating weights. Reused states are scored once, and test outcomes do not enter checkpoint or learning-rate choice.

F.7 Evaluation and uncertainty

The primary family contains supported-minus-source, supported-minus-count-only, and supported-minus-score-stratified contrasts at each of three learning rates and in each source-retrieval R@1 direction, all at epoch 10: 18 effects. For a randomized comparator, first average its three assignment draws within each training seed and query, then average the paired seed differences. The three supported fits are reused as three fits, not treated as nine independent replications.

We use 10,000 paired image-cluster bootstrap resamples with seed 20261004 and Bonferroni-adjusted percentile intervals at nominal confidence

1 – .05/18 = 99.7222% per contrast. Text-to-image resampling retains all five source-caption queries in their owning image cluster. These intervals condition on the fitted training seeds and assignment draws; neither factor is resampled. Query clusters are resampled against their original fixed retrieval gallery; gallery candidates are not regenerated. Per-seed and per-draw cells, their means, and all declared learning rates accompany the conditional image intervals. An interval spanning zero does not establish equivalence.

COCO, relation and composition tests, source metrics other than R@1, trajectories, subgroup summaries, and selected-procedure comparisons are descriptive. Source retrieval, VE400, SugarCrepe, and SugarCrepe++ cover the declared trajectory and selected states. COCO covers all epoch-10 states and unique selected states, without requiring additional epoch-1 or epoch-5 scoring. Adapter forward computation and normalization remain float32; similarities accumulate in float64. Retrieval ties use candidate-manifest order, relation ties receive half credit, and composition triplets retain the original strict comparisons. Pre-score receipts bind code, input manifests and features, and checkpoint hashes. Per-query retrieval ranks and relevance indices, relation comparisons, and composition scores are retained for independent recomputation.

F.8 Interpretation and an unexecuted allocation design

The randomized controls test label-informed assignment beyond fixed counts and, for stratification, coarse frozen-score difficulty. They do not independently adjudicate visual truth. The directional outcomes establish behavior under the specified model and schedules; neither these outcomes nor Equation 38 identifies target redistribution as the cause of every earlier retrieval loss.

The archive also contains a separate, design-only allocation factorial. It proposes holding true positive identities fixed while changing source-versus-support image target mass and reverse-anchor group weights independently, with normalized total loss weight. This would test allocation directly. It has no implementation, fitted models, or results in this study and is outside the frozen protocol. The executed controls use only the original uniform-positive objective.

G Matched Assignment Study: Complete Results

G.1 Every terminal draw and training seed

Each row below is one epoch-ten fit. Count and Strata numbers identify assignment draws; no draw is selected for reporting. All entries are percentages.

G.2 Both complete selection strategies

G.3 Complete descriptive trajectory means

G.4 Prespecified trajectories

Rate	Comparator	$\Delta I \rightarrow T$	99.7222% interval	$\Delta T \rightarrow I$	99.7222% interval
10^{-4}	Source	-6.000	$[-9.037, -3.063]$	+0.120	$[-1.087, +1.322]$
10^{-4}	Count	-0.978	$[-3.389, +1.379]$	+1.811	$[+0.631, +2.995]$
10^{-4}	Strata	-0.744	$[-3.001, +1.436]$	+1.540	$[+0.504, +2.583]$
$3 \cdot 10^{-4}$	Source	-7.133	$[-10.367, -3.796]$	-1.320	$[-2.955, +0.381]$
$3 \cdot 10^{-4}$	Count	+1.322	$[-1.730, +4.114]$	+4.838	$[+3.332, +6.376]$
$3 \cdot 10^{-4}$	Strata	+0.122	$[-2.659, +2.815]$	+3.820	$[+2.412, +5.198]$
10^{-3}	Source	-10.833	$[-14.341, -7.330]$	-1.900	$[-3.962, +0.148]$
10^{-3}	Count	+3.789	$[+0.717, +6.922]$	+7.038	$[+5.163, +8.832]$
10^{-3}	Strata	+0.222	$[-2.591, +3.025]$	+4.084	$[+2.426, +5.804]$

Table 39: All 18 primary effects. Differences are supported minus comparator, in percentage points. Randomized comparators average all three assignment draws within each training seed before seed averaging. Intervals resample images and condition on the fitted seeds and draws.

Policy	Rate	Seed	Rel.	Edited	Both	e $I \rightarrow T$	e $T \rightarrow I$	C $I \rightarrow T$	C $T \rightarrow I$
Source	10^{-4}	17	77.99	81.01	63.93	84.90	65.74	53.98	36.95
Source	10^{-4}	29	77.62	80.93	64.31	84.60	66.12	53.82	36.97
Source	10^{-4}	43	77.65	81.04	64.39	84.60	66.06	53.52	36.97
Source	$3 \cdot 10^{-4}$	17	75.23	80.22	63.65	80.20	62.62	50.56	34.36
Source	$3 \cdot 10^{-4}$	29	75.62	79.96	63.61	82.30	63.00	50.74	34.63
Source	$3 \cdot 10^{-4}$	43	75.21	80.26	64.05	82.10	63.32	50.82	34.61
Source	10^{-3}	17	69.88	76.35	59.34	76.30	52.72	43.42	26.28
Source	10^{-3}	29	71.52	76.93	60.33	75.90	54.88	43.68	26.70
Source	10^{-3}	43	70.12	75.76	61.07	75.70	54.72	43.84	26.79
Supported	10^{-4}	17	87.64	83.96	67.90	78.40	65.96	49.02	36.54
Supported	10^{-4}	29	87.69	84.01	67.71	79.10	66.14	48.96	36.41
Supported	10^{-4}	43	87.52	83.84	67.63	78.60	66.18	49.16	36.35
Supported	$3 \cdot 10^{-4}$	17	86.54	83.26	66.93	74.90	61.08	45.06	33.38
Supported	$3 \cdot 10^{-4}$	29	86.79	83.54	66.91	74.40	62.16	45.24	33.82
Supported	$3 \cdot 10^{-4}$	43	86.71	83.25	67.63	73.90	61.74	43.46	33.70
Supported	10^{-3}	17	85.82	81.47	65.21	65.20	51.34	36.44	25.97
Supported	10^{-3}	29	85.90	81.33	65.38	65.70	52.96	37.40	26.62
Supported	10^{-3}	43	85.61	81.04	66.26	64.50	52.32	35.46	26.94

Table 40: Terminal per-seed results. e denotes the full e-ViL source-caption pool and C denotes COCO. Other endpoints follow Table 37.

Policy	Rate	Seed	Rel.	Edited	Both	e $I \rightarrow T$	e $T \rightarrow I$	C $I \rightarrow T$	C $T \rightarrow I$
Count 1	10^{-4}	17	79.20	82.53	65.21	79.20	64.14	49.76	35.73
Count 1	10^{-4}	29	78.99	82.52	65.19	79.40	64.56	50.54	35.82
Count 1	10^{-4}	43	79.38	82.96	65.52	80.10	64.54	50.54	35.64
Count 1	$3 \cdot 10^{-4}$	17	76.45	80.55	63.19	72.10	56.52	44.30	31.06
Count 1	$3 \cdot 10^{-4}$	29	76.87	80.67	63.61	72.40	56.28	44.32	30.74
Count 1	$3 \cdot 10^{-4}$	43	75.85	79.79	63.63	73.30	56.54	44.04	30.42
Count 1	10^{-3}	17	74.92	77.82	59.81	61.20	45.12	34.88	22.57
Count 1	10^{-3}	29	74.66	78.15	61.57	60.00	43.60	33.94	22.14
Count 1	10^{-3}	43	73.58	77.41	60.04	63.70	43.96	34.80	22.04
Count 2	10^{-4}	17	78.90	82.89	64.91	79.90	63.98	49.76	35.55
Count 2	10^{-4}	29	79.07	82.76	65.65	78.30	64.52	49.82	35.47
Count 2	10^{-4}	43	79.20	82.93	65.15	79.70	64.28	50.12	35.48
Count 2	$3 \cdot 10^{-4}$	17	77.19	81.19	63.28	72.10	56.84	44.10	30.64
Count 2	$3 \cdot 10^{-4}$	29	75.93	81.05	63.55	73.40	56.66	44.22	30.56
Count 2	$3 \cdot 10^{-4}$	43	75.02	80.36	62.88	72.80	56.84	45.12	30.17
Count 2	10^{-3}	17	73.68	76.93	60.35	59.00	44.36	33.36	22.26
Count 2	10^{-3}	29	74.34	77.58	60.84	62.10	45.24	34.92	22.74
Count 2	10^{-3}	43	74.51	78.22	60.67	61.60	45.22	33.60	22.54

Table 41: Terminal per-seed results. e denotes the full e-ViL source-caption pool and C denotes COCO. Other endpoints follow Table 37.

Policy	Rate	Seed	Rel.	Edited	Both	e I→T	e T→I	C I→T	C T→I
Count 3	10^{-4}	17	79.26	82.65	64.98	79.80	64.18	49.98	35.15
Count 3	10^{-4}	29	78.75	82.45	64.98	80.60	64.14	50.18	35.14
Count 3	10^{-4}	43	79.05	82.61	64.83	80.10	64.20	50.14	35.29
Count 3	$3 \cdot 10^{-4}$	17	76.63	80.89	63.80	73.40	57.28	43.80	30.99
Count 3	$3 \cdot 10^{-4}$	29	75.82	80.69	63.53	74.00	56.96	44.18	30.59
Count 3	$3 \cdot 10^{-4}$	43	75.63	80.12	63.13	74.20	57.48	44.52	30.74
Count 3	10^{-3}	17	74.08	77.11	59.87	61.70	46.16	34.00	22.62
Count 3	10^{-3}	29	73.55	76.95	60.29	61.90	46.86	34.48	21.77
Count 3	10^{-3}	43	72.76	77.38	60.58	60.90	46.00	33.82	21.76
Strata 1	10^{-4}	17	81.51	82.80	65.82	78.90	64.44	50.18	35.71
Strata 1	10^{-4}	29	81.36	82.95	66.03	78.90	64.52	50.42	35.82
Strata 1	10^{-4}	43	80.98	82.98	65.92	78.90	64.64	49.88	35.70
Strata 1	$3 \cdot 10^{-4}$	17	78.88	81.47	63.88	74.00	57.64	45.44	31.31
Strata 1	$3 \cdot 10^{-4}$	29	78.19	81.15	63.95	72.30	58.22	45.42	31.26
Strata 1	$3 \cdot 10^{-4}$	43	78.24	81.07	64.70	74.10	58.52	45.46	31.38
Strata 1	10^{-3}	17	76.90	79.23	62.94	65.20	45.86	36.06	23.48
Strata 1	10^{-3}	29	76.16	77.50	62.10	63.70	49.74	37.00	24.13
Strata 1	10^{-3}	43	75.27	79.34	62.06	63.50	47.22	35.84	23.41

Table 42: Terminal per-seed results. e denotes the full e-ViL source-caption pool and C denotes COCO. Other endpoints follow Table 37.

Policy	Rate	Seed	Rel.	Edited	Both	e I→T	e T→I	C I→T	C T→I
Strata 2	10^{-4}	17	80.71	83.17	65.84	79.90	64.72	50.30	35.74
Strata 2	10^{-4}	29	81.00	83.06	66.47	79.50	64.78	50.14	35.74
Strata 2	10^{-4}	43	80.49	83.53	66.07	80.10	64.46	49.80	35.65
Strata 2	$3 \cdot 10^{-4}$	17	78.51	82.71	65.55	73.60	57.72	44.82	31.58
Strata 2	$3 \cdot 10^{-4}$	29	78.18	82.01	65.21	75.30	57.18	45.66	31.54
Strata 2	$3 \cdot 10^{-4}$	43	77.73	81.85	65.10	73.80	58.58	45.62	31.52
Strata 2	10^{-3}	17	76.49	79.10	62.73	64.80	47.82	36.80	23.46
Strata 2	10^{-3}	29	75.84	79.55	62.37	66.70	49.52	38.30	23.83
Strata 2	10^{-3}	43	76.65	79.56	62.98	65.20	48.40	37.00	24.16
Strata 3	10^{-4}	17	81.49	82.64	65.38	79.40	64.54	50.10	35.50
Strata 3	10^{-4}	29	81.24	82.93	65.57	79.90	64.54	50.14	35.62
Strata 3	10^{-4}	43	81.38	82.95	65.76	79.50	64.34	49.88	35.46
Strata 3	$3 \cdot 10^{-4}$	17	78.48	81.51	63.53	74.80	57.60	45.50	31.18
Strata 3	$3 \cdot 10^{-4}$	29	78.14	81.25	63.72	74.30	57.64	45.82	30.96
Strata 3	$3 \cdot 10^{-4}$	43	78.60	81.43	63.72	76.30	57.46	45.68	31.30
Strata 3	10^{-3}	17	76.59	79.06	61.05	63.70	47.60	36.26	23.56
Strata 3	10^{-3}	29	76.18	77.95	62.10	66.30	48.70	37.30	23.81
Strata 3	10^{-3}	43	76.86	79.11	62.14	65.10	48.24	35.02	24.02

Table 43: Terminal per-seed results. e denotes the full e-ViL source-caption pool and C denotes COCO. Other endpoints follow Table 37.

Selector	Policy	Seed	Ep.	Rel.	Edited	Both	e I→T	e T→I	C I→T	C T→I
Native	Source	17	1	81.35	83.44	67.65	84.90	69.68	56.28	39.65
Native	Source	29	1	81.09	83.42	68.09	85.00	69.74	56.02	39.46
Native	Source	43	3	81.12	82.91	66.47	84.20	68.64	55.26	39.02
Native	Supported	17	5	86.75	83.38	67.58	78.20	63.62	48.36	35.40
Native	Supported	29	10	86.79	83.54	66.91	74.40	62.16	45.24	33.82
Native	Supported	43	5	87.46	83.25	67.19	78.40	64.12	48.28	35.30
Native	Count 1	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 1	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 1	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 2	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 2	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 2	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 3	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 3	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 3	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 1	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35

Table 44: Every selected training-seed state. Each random draw has its own learning-rate and epoch selection; there is no winning-draw search. The source manifest records exact state IDs and learning rates.

Selector	Policy	Seed	Ep.	Rel.	Edited	Both	e I→T	e T→I	C I→T	C T→I
Native	Strata 1	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 1	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 2	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 2	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 2	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 3	17	7	80.11	82.09	65.00	78.30	60.84	48.36	33.71
Native	Strata 3	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 3	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Source	17	1	81.35	83.44	67.65	84.90	69.68	56.28	39.65
Retrieval	Source	29	2	80.96	83.29	67.06	84.90	69.64	56.04	39.60
Retrieval	Source	43	2	81.15	83.14	67.08	84.70	69.78	55.94	39.40
Retrieval	Supported	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Supported	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Supported	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 1	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 1	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35

Table 45: Every selected training-seed state. Each random draw has its own learning-rate and epoch selection; there is no winning-draw search. The source manifest records exact state IDs and learning rates.

Selector	Policy	Seed	Ep.	Rel.	Edited	Both	e I→T	e T→I	C I→T	C T→I
Retrieval	Count 1	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 2	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 2	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 2	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 3	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 3	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 3	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 1	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 1	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 1	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 2	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 2	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 2	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 3	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 3	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 3	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35

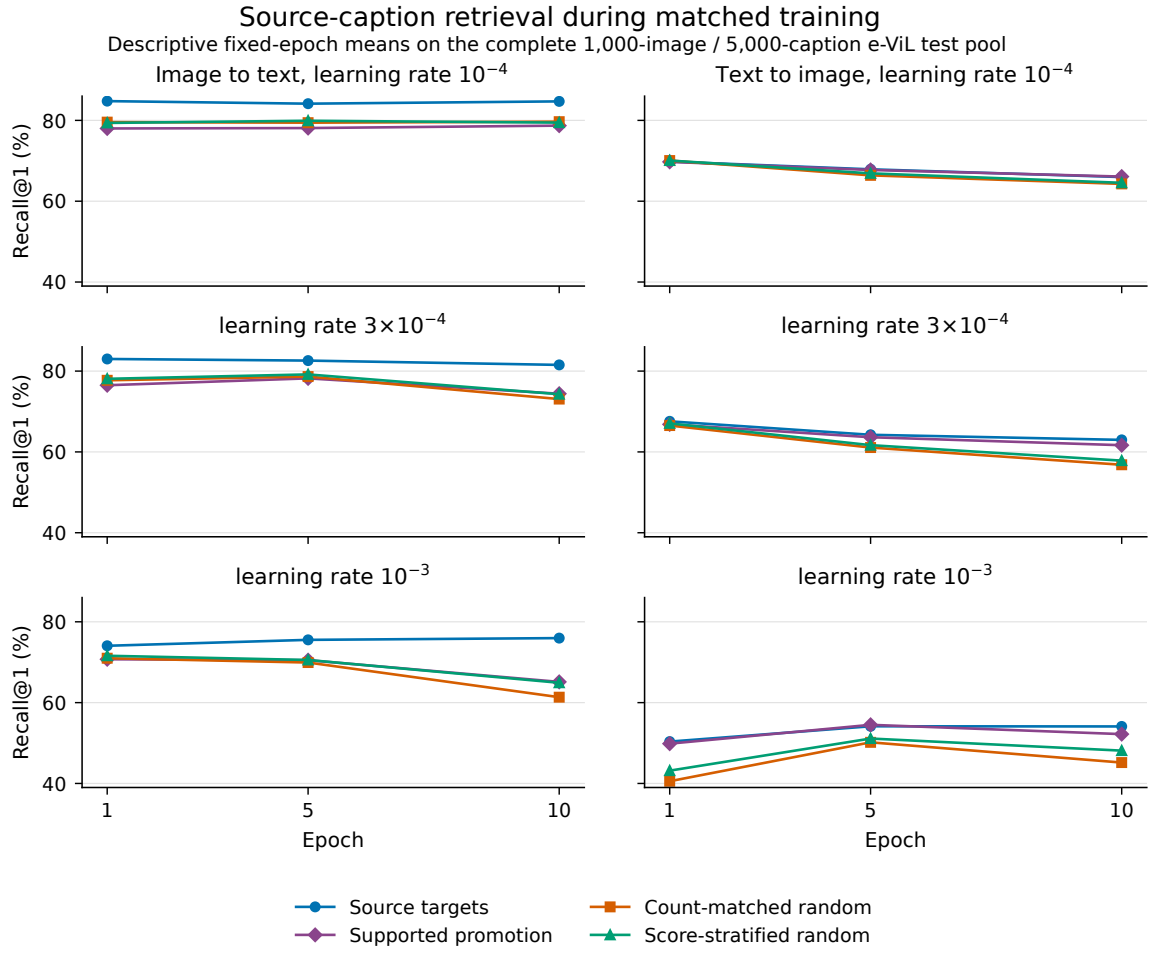
Table 46: Every selected training-seed state. Each random draw has its own learning-rate and epoch selection; there is no winning-draw search. The source manifest records exact state IDs and learning rates.

Policy	Rate	Ep.	Rel.	Edited	Both	First	Alternative	e I→T	e T→I
Source	10^{-4}	1	81.23	83.40	67.94	80.95	73.10	84.77	69.81
Source	10^{-4}	5	80.07	81.89	65.55	80.16	70.25	84.13	67.89
Source	10^{-4}	10	77.75	81.00	64.21	80.16	68.78	84.70	65.97
Source	$3 \cdot 10^{-4}$	1	80.63	82.32	65.48	80.22	70.38	83.00	67.57
Source	$3 \cdot 10^{-4}$	5	76.53	80.17	63.55	79.41	68.38	82.60	64.24
Source	$3 \cdot 10^{-4}$	10	75.35	80.14	63.77	79.22	68.63	81.53	62.98
Source	10^{-3}	1	74.94	77.38	61.03	77.34	65.76	74.07	50.36
Source	10^{-3}	5	73.25	77.57	61.71	77.35	67.05	75.53	54.15
Source	10^{-3}	10	70.51	76.35	60.25	76.59	65.51	75.97	54.11
Supported	10^{-4}	1	86.72	85.43	70.00	81.18	75.69	78.00	69.74
Supported	10^{-4}	5	87.91	84.58	68.25	80.76	73.58	78.10	67.73
Supported	10^{-4}	10	87.62	83.93	67.75	80.23	73.31	78.70	66.09
Supported	$3 \cdot 10^{-4}$	1	87.48	84.82	68.47	80.42	74.13	76.50	66.83
Supported	$3 \cdot 10^{-4}$	5	87.04	83.34	67.33	79.81	72.90	78.20	63.63
Supported	$3 \cdot 10^{-4}$	10	86.68	83.35	67.16	79.45	72.84	74.40	61.66
Supported	10^{-3}	1	86.17	81.61	65.42	78.12	71.93	70.73	49.85
Supported	10^{-3}	5	86.03	81.66	66.97	78.01	73.22	70.50	54.51
Supported	10^{-3}	10	85.78	81.28	65.62	77.37	71.94	65.13	52.21

Table 47: All declared descriptive epoch means. First and Alternative are the separate valid-caption SugarCrepe++ accuracies. Random families average all three assignments equally within seed.

Policy	Rate	Ep.	Rel.	Edited	Both	First	Alternative	e I→T	e T→I
Count mean	10^{-4}	1	83.71	84.56	69.80	80.83	75.53	79.57	70.06
Count mean	10^{-4}	5	81.39	83.49	66.13	79.83	71.24	79.43	66.40
Count mean	10^{-4}	10	79.09	82.70	65.16	79.62	70.56	79.68	64.28
Count mean	$3 \cdot 10^{-4}$	1	82.61	83.78	66.97	79.60	72.37	77.72	66.52
Count mean	$3 \cdot 10^{-4}$	5	77.88	81.98	64.02	79.18	69.17	78.64	61.08
Count mean	$3 \cdot 10^{-4}$	10	76.15	80.59	63.40	77.89	69.25	73.08	56.82
Count mean	10^{-3}	1	77.54	80.55	63.60	77.71	69.64	70.99	40.55
Count mean	10^{-3}	5	74.76	79.16	62.78	76.54	69.00	69.92	50.18
Count mean	10^{-3}	10	74.01	77.51	60.45	75.14	66.54	61.34	45.17
Strata mean	10^{-4}	1	84.19	84.61	69.84	80.82	75.73	79.37	70.06
Strata mean	10^{-4}	5	82.86	83.67	66.90	80.06	72.19	79.91	66.89
Strata mean	10^{-4}	10	81.13	83.00	65.87	79.75	71.35	79.44	64.55
Strata mean	$3 \cdot 10^{-4}$	1	83.65	83.84	67.52	79.70	73.11	78.09	67.06
Strata mean	$3 \cdot 10^{-4}$	5	80.18	82.38	65.28	79.28	70.88	79.18	61.67
Strata mean	$3 \cdot 10^{-4}$	10	78.33	81.60	64.37	78.54	70.09	74.28	57.84
Strata mean	10^{-3}	1	79.54	80.50	64.05	77.54	70.21	71.60	43.17
Strata mean	10^{-3}	5	77.59	79.91	64.49	77.04	70.70	70.56	51.11
Strata mean	10^{-3}	10	76.32	78.93	62.27	76.27	68.27	64.91	48.12

Table 48: All declared descriptive epoch means. First and Alternative are the separate valid-caption SugarCrepe++ accuracies. Random families average all three assignments equally within seed.



Every planned learning rate and epoch is shown. Three training seeds; randomized policies average three fixed assignment draws.

Figure 3: Descriptive source-pool retrieval at epochs one, five, and ten. Every learning rate is displayed. Randomized families average all assignment draws equally within seed. The primary family concerns epoch ten only.

H Prior Study: Context and Selected Procedures

The following appendices preserve the completed twelve-policy study that motivated the matched-assignment experiment. Its fits, independently selected procedures, six-effect inferential family, and 400-image relation pool are distinct from the follow-up’s matched terminal contrasts and full source-caption retrieval pool. Its tables do not estimate the effect of positive assignment at a common training schedule. All source files and results of that study are retained unchanged in the project archive; this appendix changes only presentation and cross-references.

The selected-procedure comparison used one common tuning budget per rule. The complete policy definitions, source provenance, per-seed outcomes, diagnostic, and mathematical analysis fol-



Source positives: A young girl is preparing to blow out her candle.

Positive expansion: a woman is making some food in a bowl

Figure 4: Top retrieved captions for the same image query (seed 17). The best annotated caption moves from rank 1 to 6. Both models search the full caption pool.

low. The caption-retention figure below concerns only the original selected states; the follow-up retains additional early states and reports their trajectories separately.

I Prior Study: Complete Policy Definitions

A batch has B unique image rows and T text columns, containing all selected source and hypothesis captions for its images. The relation matrix $R \in \{0, 1, 2, 3, 4\}^{B \times T}$ uses codes for unannotated, source, supported, contradicted, and neutral pairs, respectively. Relations apply only to their annotated image–text pair. An absent cross-image annotation is an eligible negative, not a neutral label.

Let J denote the candidate set for a row or column, $P \subseteq J$ its positive set, and P_{ij} the binary positive mask. Every contrastive policy includes source captions. Expanded policies additionally include supported hypotheses. Candidate weights start at one and are changed according to Table 52; all positive weights are then forced to one. The scalar loss averages eligible image rows and eligible text columns independently, with weight $1/2$ on each direction. The reverse direction transposes the same mask and weight matrices. Columns without a positive image are omitted from the reverse average. These rules preserve the meaning of rectangular batches with several captions per image.

Detached similarity weights. Let t_j be the unit frozen text feature for candidate j , and C_i the

Policy	e-SNLI-VE	SugarCrepe	SC++ both	COCO I→T	COCO T→I
Frozen	79.47	81.21	68.95	56.32	39.35
Source positives	81.18 $\pm$ 0.14	83.26 $\pm$ 0.30	67.40	55.85	39.38
Positive expansion	87.00 $\pm$ 0.40	83.39 $\pm$ 0.15	67.23	47.29	34.84
+ contradiction weighting	87.28 $\pm$ 0.33	83.33 $\pm$ 0.16	67.30	47.25	34.81
+ neutral exclusion	86.66 $\pm$ 0.44	83.39 $\pm$ 0.27	67.66	48.01	35.39
+ both	86.74 $\pm$ 0.17	83.24 $\pm$ 0.31	67.61	47.26	35.27
Constant weights	82.15 $\pm$ 0.24	83.67 $\pm$ 0.20	68.37	55.21	39.39

Table 49: Relation interventions with the same candidate pool and tuning budget. Scores are percentages; subscripts show sample standard deviation across three selected seeds for the primary endpoints. COCO columns report full-pool R@1. The three rows marked + add rules to positive expansion. Constant weights use source positives.

Edit group	n	Source	Expand.	Δ
Added content	2,754	88.09	90.01	+1.92
Replace/swap	4,757	80.46	79.56	-0.90
All edits	7,511	83.26	83.39	+0.14

Table 50: SugarCrepe accuracy by edit group. Differences are positive expansion minus source positives, in percentage points. Groups partition the official pairs.

Weights	e-SNLI-VE	Sugar	I→T
Fixed similarity	79.56	81.80	54.33
Shuffled weights	83.34	83.49	52.27
Median similarity	82.66	83.73	53.97

Table 51: Similarity weights and their shuffle, using source positives. Entries are mean percentages; the final column is COCO R@1. The constant-weight reference appears in Table 49.

source-caption indices of image i . Define

$$a_{ij} = \frac{1 + |C_i|^{-1} \sum_{k \in C_i} t_k^\top t_j}{2}. \quad (39)$$

The average of source features is not renormalized. The negative weight is

$$w_{ij} = \text{sigmoid}\left(\frac{\tau - a_{ij}}{0.05}\right). \quad (40)$$

The fixed rule sets $\tau = 0.5$. The median rule uses the statistical median of a_{ij} over nonpositive cells of the current batch, averaging the two central values for an even count. Positive weights remain one. All features, thresholds, and weights in this calculation are detached from the training graph. The kernel never uses adapted features.

Shuffling and random exclusion. The shuffled rule permutes the complete batch-wide multiset of negative weights from the fixed-threshold rule, leaving positive weights intact. It preserves the weight distribution but need not preserve each row’s weighted mass. Random exclusion samples nonpositive cells without replacement within each image row, matching that image’s number of neutral annotations. Because the reverse direction uses a transpose, its per-text exclusion counts need not match the neutral rule. Both randomizations use a seeded generator whose state advances through batches and epochs.

Smoothing and ranking. Smoothing uses source-positive targets and unit weights. Its row target is $0.9 \text{Unif}(P) + 0.1 \text{Unif}(J)$, where J contains every candidate. For ranking, let E_i and C'_i denote supported and contradicted hypotheses of image i . The eligible-image loss is

$$\ell_i^{\text{rank}} = \frac{\sum_{e \in E_i} \sum_{c \in C'_i} \text{softplus}(s_{ic} - s_{ie})}{|E_i| |C'_i|}. \quad (41)$$

We average over images with both sets nonempty. Ranking has no reverse direction and does not train on source, neutral, or unannotated pairs.

J Prior Study: Data, Provenance, and Reconstruction

The current execution was reconstructed from retained methods and proof text after the earlier workspace became unavailable. Historical aggregate tables are kept in a separate context directory and are not inputs to training, evaluation, or paper tables. The exact old similarity kernel and the base target set for smoothing were not recoverable. The explicit definitions in Appendix I were fixed

Archive name	Positive set	Weight or target rule
clip	Source	All weights one
multipositive	Source + supported	All weights one
grounded	Source + supported	Contradictions: 2; neutrals: 0
grounded_no_hardenening	Source + supported	Neutrals: 0
grounded_no_abstention	Source + supported	Contradictions: 2
random_exclusion	Source + supported	Count-matched random exclusions
smoothing	Source	$0.9 \text{ Unif}(P) + 0.1 \text{ Unif}(J)$
constant	Source	All negatives: 0.25
sanw_fixed	Source	Similarity kernel, threshold 0.5
sanw_median	Source	Similarity kernel, batch median threshold
shuffled	Source	Permuted fixed-threshold negative weights
pairwise_rank	Supported/contradicted	Image-balanced pairwise softplus

Table 52: All twelve trainable policies. A weight of zero exactly excludes a pair. The frozen reference has no trained parameters.

Split	Images	Support	Contradict	Neutral
Train	1,200	5,319	5,846	5,027
Calibration	100	522	554	355
Validation	100	502	534	339
Test	400	2,088	2,278	1,495

Table 53: Retained hypothesis annotations. Each image also has five source captions. All validation and test images, and 1,192 of 1,200 training images, have both supported and contradicted hypotheses.

before the new runs and are documented in the reconstruction record. The prior twelve-policy study was a new execution under those recorded choices. The matched-assignment follow-up preserves those original inputs and outputs.

J.1 Relation annotations and source captions

We use the official e-ViL release of e-SNLI-VE at revision 94f17f5c1895. Its training, development, and test sources contain 401,717, 14,339, and 14,740 annotated hypotheses over 29,783, 1,000, and 1,000 images, respectively. The official pools have no shared image IDs or annotation-pair IDs. With Python’s `random.Random(42)`, we sample from sorted image IDs: 1,200 training images, 200 development images split into 100 calibration and 100 validation images, and 400 test images. All annotations of each selected image remain in source order. Table 53 gives the resulting counts.

The source captions come from the official Flickr30k Entities annotation archive at revision 68b3d6f12d1d. We use its five sentence-file

lines per image in stored order. Entity markup is removed while retaining source words, case, and punctuation, with single spaces between stored tokens. This source covers all 31,783 images. An incomplete alternative caption mirror covers only 31,014 and was not used. The image bytes come from the `nlphuji/flickr30k` mirror at revision 2b239befc81b. Each selected image passes ZIP size/CRC checks, JPEG verification, and full decoding. No selected image is silently skipped.

An earlier deprecated relation release was rejected during an input-only audit because of development/test overlap. This decision preceded feature-based evaluation. The chosen official release, exact IDs, source URLs, full revisions, file hashes, and selection order are recorded in the per-dataset provenance files. The released labels include inherited text-inference judgments, so their names are operational annotation categories rather than a claim of new visual adjudication.

J.2 Transfer benchmarks

SugarCrepes uses all 7,511 pairs over 1,560 images from revision 0047054b2439. Its category counts are 692 added attributes, 2,062 added objects, 788 replaced attributes, 1,652 replaced objects, 1,406 replaced relations, 666 swapped attributes, and 245 swapped objects. SugarCrepes++ uses all 4,757 triplets over 1,542 images from revision c083bbf039a5; it covers the five replacement/swap categories with the same category counts.

Each benchmark preserves its own released text

strings. SugarCrepe++ changes 1,203 first-positive strings and 878 foil strings relative to the matched SugarCrepe records. Its first-positive accuracy therefore cannot substitute for SugarCrepe’s replacement/swap score. Two released SugarCrepe++ triplets have identical first-positive and foil strings, making a strict win impossible; these remain in the official denominator. One triplet repeats the same string in both positive positions. No item is edited or removed in response to model scores.

COCO uses all 5,000 Stanford Karpathy test images. The source contains 25,010 captions; taking the first five in stored order for every image yields 25,000 candidates and omits ten sixth captions. Caption strings are not normalized. The COCO JPEG cache contains 6,377 distinct benchmark images. It uses the official `val2014` source for Karpathy members and `val2017` for remaining benchmark images. All 183 shared SugarCrepe/Karpathy images were checked across these source versions and are byte-identical.

J.3 Overlap and duplicate audit

All 1,542 SugarCrepe++ images also occur in SugarCrepe. COCO retrieval shares 183 images with SugarCrepe and 182 with SugarCrepe++. These are complementary measurements on overlapping pools. The selected Flickr30k images share neither IDs nor image-byte hashes with the COCO-derived benchmarks. The four e-SNLI-VE splits also have no shared image bytes.

The input audit finds 48 repeated exact image/text groups in e-SNLI-VE, each appearing twice. Of these, 46 are repeated hypothesis groups. Their relation labels do not conflict. All rows are retained, so repeated annotations retain their multiplicity in the loss and metrics. The two positive/foil equality cases account for the within-triplet relation conflicts in SugarCrepe++. The audit records source defects before scoring rather than changing benchmark contents.

J.4 Frozen feature extraction

The encoder is laion/CLIP-ViT-B-32-laion2B-s34B-b79K, revision 1a25a446712b, with weight SHA256 beginning ac4f8c4b88af. We use OpenCLIP 2.32.0 and PyTorch 2.7.1 on CPU. Images are resized with bicubic interpolation, center-cropped to 224×224 , converted to RGB, and normalized with the checkpoint’s channel means and standard deviations. The 512-dimensional image and text outputs

are stored in float32 after unit L2 normalization. The fixed scale is 100.00000762939453.

Text encoding trims only causal padding positions after the last end-of-text token in a batch. A parity check compares normalized outputs against the stock encoder with tolerance $2 \cdot 10^{-6}$. Recorded per-dataset parity checks, preprocessing parameters, source hashes, input-manifest hashes, feature-cache hashes, and full model pins are stored in each feature metadata file. Reused features require exact image-byte or text-string identity under the same encoder and preprocessing.

K Prior Study: Training and Checkpoint Selection

Let $x, z \in \mathbb{R}^{512}$ be unit frozen image and text features. The trainable maps are

$$\hat{x} = \frac{x + A_I x}{\|x + A_I x\|_2}, \quad \hat{z} = \frac{z + A_T z}{\|z + A_T z\|_2}. \quad (42)$$

Both 512×512 matrices start at zero and have no bias. Their total parameter count is $2 \cdot 512^2 = 524,288$. The score is $s_{ij} = \alpha \hat{x}_i^\top \hat{z}_j$, where $\alpha > 0$ is the pretrained checkpoint’s fixed scale. Feature extraction and source hashes identify the exact encoder and input order.

The learning-rate grid is $\{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}\}$ and training seeds are 17, 29, 43. Each candidate trains for ten epochs with image batches of 32. All source captions and hypotheses assigned to those images enter the text candidate pool. AdamW uses weight decay 0.01, betas (0.9, 0.999), and epsilon 10^{-8} . The gradient norm is clipped at one. There is no warmup or learning-rate schedule. Separate random generators control image order and stochastic policy operations, so weight shuffling does not alter batch order.

Validation retrieval uses every validation caption and hypothesis as a candidate. Both source captions and supported hypotheses are relevant matches. Its R@1 and the image-balanced supported-versus-contradicted accuracy receive equal weight in the selection score. The epoch with the highest score is retained for each seed and learning rate, including epoch zero. Exact ties retain the earliest epoch. A policy selects one learning rate by averaging these best scores over all three seeds; exact ties select the smaller rate. Held-out scoring starts after all 108 candidates and 36 selections are frozen.

Ranking evaluation accumulates dot products in float64 over adapted float32 normalized features.

Exact retrieval-score ties follow candidate manifest order; relation-ranking ties receive half credit. The fixed positive score scale is omitted from ranking metrics because it cannot change the order. Images without both supported and contradicted hypotheses are omitted only from the relation average, with their count recorded.

An immutable execution ledger binds the protocol, code, manifest, feature metadata, cache, and split IDs before fitting. Every candidate saves its best state, all eleven validation evaluations, and a completion record with artifact hashes. The root epoch-zero state is retained separately. Resume accepts a completed candidate only when its identities and hashes match. Interrupted candidates restart deterministically from epoch zero. Selected checkpoints are copied into a separate tree while all candidate states remain available.

L Prior Study: Metrics and Statistical Comparisons

For e-SNLI-VE, compare every supported hypothesis against every contradicted hypothesis within an image. A larger supported score receives one point and an exact tie receives half a point. Average pairs within each eligible image, then average images. This gives each image equal weight despite different numbers of annotated statements.

SugarCrepe uses a strict positive-score comparison against its foil. SugarCrepe++ reports each positive separately and their conjunction against the same foil. The conjunction is one only if both scores are strictly greater. Per-category results retain official labels. The addition group pools added objects and attributes; the combined replacement/swap group pools the remaining categories. Grouped scores use item counts as weights, not an unweighted mean of category percentages.

COCO image-to-text R@1 counts an image query as correct when its highest-scoring caption belongs to the image’s annotated caption set. Text-to-image R@1 requires the source image at rank one. All candidates remain in the full pool during scoring. Source-caption ownership is the operational relevance label.

The six primary contrasts compare the combined relation policy against positive expansion, neutral exclusion alone, and contradiction weighting alone, on e-SNLI-VE and SugarCrepe. For each contrast, average the paired outcomes across the three selected seeds before sampling images. Each boot-

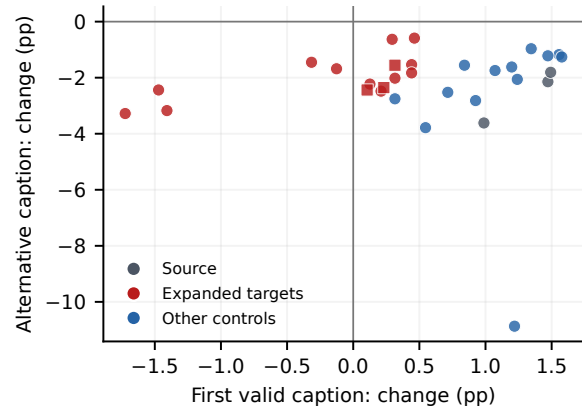


Figure 5: Changes on the same SugarCrepe++ triplets relative to frozen. Each point is one selected nonzero-epoch adapter. All 31 original selected adapters lose accuracy on the alternative valid caption; 26 improve on the first. The 5 epoch-zero selections are omitted.

strap draw samples images with replacement and retains all associated items. With 10,000 replicates, seed 20261003, and family size six, two-sided percentile bounds use tail probability $0.05/(2 \cdot 6)$, yielding 99.1667% intervals. They condition on the selected runs and data images. Secondary metrics and category groupings are descriptive.

M Prior Study: Retention Across Selected Adapters

The first-positive and alternative-positive scores use the same triplets and foil. The decline in alternative-caption accuracy is relative to frozen and is not specific to positive expansion. Positive expansion has slightly higher alternative-caption accuracy than source-positive adaptation. Full per-policy and per-seed values follow below.

N Prior Study: Full Fresh Results and Execution Checks

Tables 54–61 give all selected-policy means, seed outcomes, primary intervals, category scores, and retrieval transitions. Five selected states retain epoch zero: all three ranking runs and fixed-similarity seeds 29 and 43. Those states have predictions identical to the frozen reference.

The illustrative query in Figure 4 is coco 60623. It is the first image in COCO manifest order that source-positive adaptation retrieves correctly at rank one in all three seeds, while positive expansion fails at rank one in every seed and falls outside the top five in seed 17. The displayed captions are the two seed-17 top results. This deterministic post

Policy	e-SNLI-VE	SugarCrepe	SC++ both
Frozen	79.47	81.21	68.95
Source	81.18 $\pm$ 0.14	83.26 $\pm$ 0.30	67.40 $\pm$ 0.84
Expanded	87.00 $\pm$ 0.40	83.39 $\pm$ 0.15	67.23 $\pm$ 0.34
Contradiction	87.28 $\pm$ 0.33	83.33 $\pm$ 0.16	67.30 $\pm$ 0.28
Neutral	86.66 $\pm$ 0.44	83.39 $\pm$ 0.27	67.66 $\pm$ 0.52
Combined	86.74 $\pm$ 0.17	83.24 $\pm$ 0.31	67.61 $\pm$ 0.45
Random	86.26 $\pm$ 0.28	81.62 $\pm$ 0.70	65.93 $\pm$ 0.57
Smoothing	79.23 $\pm$ 4.85	82.28 $\pm$ 2.18	65.97 $\pm$ 4.59
Ranking	79.47 $\pm$ 0.00	81.21 $\pm$ 0.00	68.95 $\pm$ 0.00
Fixed	79.56 $\pm$ 0.15	81.80 $\pm$ 1.01	68.00 $\pm$ 1.65
Shuffled	83.34 $\pm$ 0.57	83.49 $\pm$ 0.36	66.90 $\pm$ 0.26
Median	82.66 $\pm$ 0.19	83.73 $\pm$ 0.17	68.01 $\pm$ 0.07
Constant	82.15 $\pm$ 0.24	83.67 $\pm$ 0.20	68.37 $\pm$ 0.48

Table 54: All selected policies on the three discrimination endpoints. Entries are percentages with sample seed standard deviations. Source, Expanded, Contradiction, Neutral, and Combined correspond to the first five trainable policies in Table 49. Other names follow Table 52.

hoc choice illustrates a recorded failure and is not a representative sample or an extra significance test.

O Separate Relation-Estimator Diagnostic

This diagnostic predicts released supported, contradicted, and neutral labels from frozen features. It does not provide labels or weights to the adapter study. Source captions and unannotated cross-image pairs are excluded. Each annotated image/hypothesis pair is one example, and each image remains entirely within its training, validation, calibration, or test split.

Features and fitting. The image-only and text-only models use the corresponding 512-dimensional features. The multimodal model concatenates image, text, their elementwise product, and their absolute difference. A standard scaler is fitted only on training rows. Multinomial logistic regression uses an intercept, L2 regularization, `lbfgs`, tolerance 10^{-6} , no class weights, and seed 17. The grid $C \in \{0.01, 0.1, 1, 10\}$ is selected by validation macro-F1, with smaller C breaking ties. There is no refit on validation data.

The original reconstruction set a 2,000-iteration cap. Text fitting reached that cap, so a separately recorded amendment raised the cap to 10,000 without changing the model, grid, splits, or tolerance. The amended run has its own ledger and output directory; the original diagnostic remains archived.

This numerical-completion amendment does not alter the frozen primary-adapter protocol.

Calibration and selection gates. Sort calibration image IDs and permute them with NumPy seed zero. The first 50 images fit a positive scalar temperature by minimizing mean three-class negative log likelihood; the other 50 select gates. Temperature lies in $[0.05, 20]$, optimized in log space, with one and both endpoints also considered explicitly. It preserves the classifier’s argmax predictions.

For supported and contradicted predictions separately, a gate requires that class to be the argmax and its calibrated probability to meet or exceed a threshold. Search complete confidence-tie groups and choose maximum coverage among thresholds accepting at least 20 calibration pairs with a two-sided 95% Wilson lower endpoint of at least 0.90. If no threshold qualifies, the gate accepts nothing. This calibration criterion is not a simultaneous guarantee after threshold search. Final precision and coverage are measured on held-out test images. Coverage divides accepted count by all test hypotheses; precision divides correct accepted predictions by accepted count. An empty gate has undefined precision.

Metrics and out-of-fold predictions. The diagnostic reports accuracy, three-class macro-F1, 10-bin expected calibration error, Brier score, and negative log likelihood. Brier score sums squared error over the three classes before averaging examples. Calibration bins have equal width and include

Policy	SC++ first	SC++ alternative	COCO I→T	COCO T→I
Frozen	79.46	75.01	56.32	39.35
Source	80.78 $\pm$ 0.29	72.48 $\pm$ 0.96	55.85 $\pm$ 0.53	39.38 $\pm$ 0.32
Expanded	79.68 $\pm$ 0.11	72.89 $\pm$ 0.49	47.29 $\pm$ 1.78	34.84 $\pm$ 0.89
Contradiction	79.72 $\pm$ 0.16	72.92 $\pm$ 0.49	47.25 $\pm$ 1.94	34.81 $\pm$ 0.92
Neutral	79.72 $\pm$ 0.33	73.64 $\pm$ 0.68	48.01 $\pm$ 0.15	35.39 $\pm$ 0.23
Combined	79.56 $\pm$ 0.36	73.64 $\pm$ 0.70	47.26 $\pm$ 1.17	35.27 $\pm$ 0.44
Random	77.93 $\pm$ 0.17	72.04 $\pm$ 0.46	43.43 $\pm$ 1.23	29.95 $\pm$ 0.65
Smoothing	80.88 $\pm$ 0.18	70.56 $\pm$ 5.56	53.85 $\pm$ 2.54	37.08 $\pm$ 3.15
Ranking	79.46 $\pm$ 0.00	75.01 $\pm$ 0.00	56.32 $\pm$ 0.00	39.35 $\pm$ 0.00
Fixed	79.64 $\pm$ 0.32	73.74 $\pm$ 2.18	54.33 $\pm$ 3.45	36.86 $\pm$ 4.31
Shuffled	80.11 $\pm$ 0.31	72.31 $\pm$ 0.15	52.27 $\pm$ 0.29	38.19 $\pm$ 0.12
Median	80.50 $\pm$ 0.18	73.37 $\pm$ 0.10	53.97 $\pm$ 0.40	39.21 $\pm$ 0.27
Constant	80.84 $\pm$ 0.16	73.60 $\pm$ 0.58	55.21 $\pm$ 0.14	39.39 $\pm$ 0.09

Table 55: Valid-caption and retrieval retention. Entries are percentages with sample seed standard deviations.

probability one in the last bin. Class ties follow the fixed order supported, contradicted, neutral. Nonempty gate precision receives a descriptive 95% image-bootstrap interval from 10,000 replicates with seed 20261003. Empty-acceptance replicates are counted and omitted from the percentile calculation.

For reusable training predictions, sort training image IDs, permute with seed 17, and divide them into three image-disjoint folds. Each fold trains on the other two folds, selects C on the external validation split, and calibrates on the external calibration halves. Thus held-out training labels do not enter the model or selection decisions producing their predictions. These predictions are archived for future work and do not enter adapter fitting.

P Prior Study: Reproducibility and Extensions

The archive separates source acquisition, normalized feature extraction, fit/selection, held-out prediction, statistical analysis, and manuscript generation. Manifests give stable IDs and input hashes. Candidate checkpoints and validation histories establish which states were eligible for selection. Selected predictions support metric reconstruction without rerunning training. Result tables are generated from the fresh analysis artifact rather than entered from the historical manuscript.

Extensions retained for future work include a second backbone, full-encoder training, more training data and seeds, independently image-adjudicated relations, denominator-matched and exact gradient-

mass-matched training controls, hardness-matched exclusion, the summed-positive objective, predicted relations, inverse-multiplicity weighting, retention-constrained selection, feature distillation, many-to-many retrieval relevance, natural deployment queries, and frozen retrieval followed by relation-adapter reranking. A reranker can change the top result even when its candidate generator is fixed. Matching controls must specify the image and text directions separately. These are distinct from the completed comparisons reported in the fresh results.

Q Prior Study: Analytical Identities and Their Scope

This appendix preserves the prior study’s positive-target identities and complete negative-weight analysis. Unless an alternative is specified, the statements concern the unsmoothed uniform-positive objective at specified scores. Their proofs do not depend on experimental outcomes.

Q.1 Notation and the common gradient

Let J be a finite candidate set, $P \subseteq J$ a nonempty positive set, and $N = J \setminus P$. Each score $s_j \in \mathbb{R}$ is finite. Weights $w_j \geq 0$ are fixed for differentiation and satisfy $w_p = 1$ for $p \in P$. Define

$$D = \sum_{j \in J} w_j e^{s_j}, \quad q_j = \frac{w_j e^{s_j}}{D}, \quad (43)$$

$$y_j = \frac{\mathbf{1}\{j \in P\}}{|P|}, \quad \ell = \log D - \sum_j y_j s_j. \quad (44)$$

Policy	Seed	Rate	Epoch	e-SNLI	Sugar	SC++	I→T	T→I
Source	17	0.0001	1	81.35	83.44	67.65	56.28	39.65
Source	29	0.0001	1	81.09	83.42	68.09	56.02	39.46
Source	43	0.0001	3	81.12	82.91	66.47	55.26	39.02
Expanded	17	0.0003	5	86.75	83.38	67.58	48.36	35.40
Expanded	29	0.0003	10	86.79	83.54	66.91	45.24	33.82
Expanded	43	0.0003	5	87.46	83.25	67.19	48.28	35.30
Contradiction	17	0.0003	5	87.04	83.33	67.58	48.46	35.48
Contradiction	29	0.0003	10	87.13	83.49	67.02	45.02	33.76
Contradiction	43	0.0003	5	87.66	83.17	67.31	48.28	35.18
Neutral	17	0.0003	5	86.42	83.60	68.26	48.10	35.54
Neutral	29	0.0003	5	86.40	83.49	67.35	47.84	35.50
Neutral	43	0.0003	5	87.17	83.09	67.37	48.10	35.12
Combined	17	0.0003	5	86.64	83.46	68.09	48.04	35.53
Combined	29	0.0003	5	86.65	83.37	67.19	47.82	35.51
Combined	43	0.0003	7	86.94	82.88	67.56	45.92	34.77
Random	17	0.001	3	86.04	80.96	65.48	44.84	29.33
Random	29	0.001	5	86.17	81.55	66.58	42.80	30.62
Random	43	0.001	3	86.58	82.36	65.73	42.64	29.90

Table 56: Individual selected checkpoints. SC++ requires both positives to beat the foil; retrieval columns are full-pool R@1. All metric entries are percentages.

Since P is nonempty and scores are finite, $D > 0$. A zero weight removes a candidate from the denominator, while its target is also zero.

Lemma 1 (Fixed-weight row gradient). *For each $j \in J$, $\partial\ell/\partial s_j = q_j - y_j$.*

Proof. Differentiating the finite sum gives $\partial D/\partial s_j = w_j e^{s_j}$. The logarithm contributes $w_j e^{s_j}/D = q_j$, and the linear target term contributes $-y_j$. $\square$

An individual positive derivative need not be negative. If $P = \{1, 2\}$ and $q = (0.8, 0.1, 0.1)$, the gradient is $(0.3, -0.4, 0.1)$. However,

$$\sum_{p \in P} \frac{\partial \ell}{\partial s_p} = - \sum_{n \in N} q_n \leq 0. \quad (45)$$

Thus claims about total positive attraction do not imply a sign for every positive score.

Q.2 Expansion of the positive set

Proposition 2 (Target redistribution). *Assume unit weights on every candidate and fixed scores. Let P contain $K \geq 1$ positives. Relabel a disjoint set $A \subseteq J \setminus P$ of $M \geq 0$ already-present candidates as positive, yielding $P' = P \cup A$. For $p \in P$,*

$$\frac{\partial \ell'}{\partial s_p} - \frac{\partial \ell}{\partial s_p} = \frac{M}{K(K+M)}. \quad (46)$$

For $a \in A$, the difference is $-1/(K+M)$. For the remaining candidates, it is zero.

Proof. The scores, candidates, and weights are unchanged, so D and all q_j remain fixed. For an original positive, subtracting the two derivatives gives

$$(q_p - 1/(K+M)) - (q_p - 1/K) = \frac{M}{K(K+M)}. \quad (47)$$

A newly positive candidate changes target probability from zero to $1/(K+M)$. The remaining targets stay zero. The original positives jointly lose $M/(K+M)$ target probability, precisely the amount assigned to the new positives. $\square$

The premise that candidates are already present matters: adding new candidates changes the denominator as well. Likewise, changing weights changes q . The following identity separately accounts for a change in the eligible text anchors of symmetric training.

Equation 38 gives the corresponding reverse-anchor averaging identity and its assumptions.

Policy	Seed	Rate	Epoch	e-SNLI	Sugar	SC++	I→T	T→I
Smoothing	17	0.0001	1	82.04	83.61	68.59	55.26	38.99
Smoothing	29	0.0001	10	73.63	79.76	60.67	50.92	33.45
Smoothing	43	0.0001	1	82.03	83.48	68.64	55.36	38.81
Ranking	17	0.0001	0	79.47	81.21	68.95	56.32	39.35
Ranking	29	0.0001	0	79.47	81.21	68.95	56.32	39.35
Ranking	43	0.0001	0	79.47	81.21	68.95	56.32	39.35
Fixed	17	0.0003	6	79.74	82.97	66.09	50.34	31.88
Fixed	29	0.0003	0	79.47	81.21	68.95	56.32	39.35
Fixed	43	0.0003	0	79.47	81.21	68.95	56.32	39.35
Shuffled	17	0.0003	1	82.85	83.10	66.64	52.60	38.10
Shuffled	29	0.0003	1	83.97	83.81	66.89	52.14	38.14
Shuffled	43	0.0003	1	83.20	83.54	67.16	52.06	38.33
Median	17	0.0001	3	82.76	83.54	67.96	53.64	38.90
Median	29	0.0001	2	82.44	83.74	67.96	54.42	39.42
Median	43	0.0001	2	82.78	83.89	68.09	53.86	39.30
Constant	17	0.0001	1	82.00	83.88	68.59	55.36	39.48
Constant	29	0.0001	1	82.03	83.65	68.70	55.10	39.39
Constant	43	0.0001	2	82.43	83.48	67.82	55.16	39.31

Table 57: Individual selected checkpoints. SC++ requires both positives to beat the foil; retrieval columns are full-pool R@1. All metric entries are percentages.

Comparator	Endpoint	Δ	99.17% interval	Seed 17	Seed 29	Seed 43
Expanded	e-SNLI-VE	-0.255	[-0.605, +0.065]	-0.109	-0.137	-0.520
Expanded	SugarCrepe	-0.155	[-0.509, +0.209]	+0.080	-0.173	-0.373
Neutral	e-SNLI-VE	+0.081	[-0.100, +0.273]	+0.221	+0.253	-0.230
Neutral	SugarCrepe	-0.155	[-0.371, +0.056]	-0.133	-0.120	-0.213
Contradiction	e-SNLI-VE	-0.535	[-0.890, -0.193]	-0.398	-0.484	-0.723
Contradiction	SugarCrepe	-0.093	[-0.442, +0.261]	+0.133	-0.120	-0.293

Table 58: Six primary contrasts: Combined minus the comparator, in percentage points. Intervals use paired image-cluster resampling and the fixed six-comparison Bonferroni family.

Q.3 A constant matching the normalizer

Proposition 3 (Matched negative mass). *Let N be nonempty under the common assumptions. Define*

$$c^* = \frac{\sum_{n \in N} w_n e^{s_n}}{\sum_{n \in N} e^{s_n}}. \quad (48)$$

Compute c^ at the specified scores and hold it fixed for differentiation. Replace w_n by this constant for each $n \in N$, keeping positive weights one. At these scores, the new normalizer, loss, each positive probability and derivative, and total negative derivative equal their original values.*

Proof. The denominator defining c^* is positive because N is nonempty and every exponential is positive. Therefore c^* exists, is nonnegative, and satis-

fies

$$\sum_{n \in N} c^* e^{s_n} = \sum_{n \in N} w_n e^{s_n}. \quad (49)$$

The positive exponential mass is unchanged. Adding it to both sides proves equality of the normalizers. The target term also stays fixed, giving equal losses. Each positive probability is e^{s_p}/D , so its probability and derivative are equal. Finally, the total negative derivative is the negative exponential mass divided by D in each case. $\square$

Individual negative derivatives are generally unequal: they are $w_n e^{s_n}/D$ and $c^* e^{s_n}/D$. Matching preserves the total amount of competition, while changing its allocation. This is a row-specific analytical construction; a single global constant need not match every row. If instead $c^*(s)$ is recomputed

Policy	Add A	Add O	Repl. A	Repl. O	Repl. R	Swap A	Swap O
Frozen	77.89	87.15	82.49	93.77	68.92	67.42	60.00
Source	83.24	89.72	83.76	94.09	72.00	67.47	61.77
Expanded	87.86	90.74	82.91	93.02	72.00	65.12	60.68
Contradiction	88.05	90.59	83.04	92.88	72.02	64.96	60.27
Neutral	87.91	90.98	82.91	93.18	71.46	65.17	60.41
Combined	87.91	90.74	83.04	93.20	71.08	65.27	59.05
Random	86.85	89.19	81.09	92.39	68.66	63.06	57.14
Smoothing	80.54	87.20	83.46	93.48	72.43	67.62	62.99
Ranking	77.89	87.15	82.49	93.77	68.92	67.42	60.00
Fixed	80.11	87.86	82.66	93.79	69.77	66.97	61.36
Shuffled	86.37	90.88	83.08	93.68	70.63	68.07	61.36
Median	85.12	91.21	83.71	94.05	71.03	68.22	62.31
Constant	84.20	90.80	83.46	94.23	72.00	67.57	62.31

Table 59: SugarCrep category means (percent). A, O, and R denote attribute, object, and relation. Category denominators are listed in Appendix J.

Seed	Queries	Pool	Lost	New	Net loss
17	I→T	5,000	649	253	396
17	T→I	25,000	2,195	1,132	1,063
29	I→T	5,000	832	293	539
29	T→I	25,000	2,651	1,241	1,410
43	I→T	5,000	597	248	349
43	T→I	25,000	1,976	1,045	931

Table 60: Retrieval success transitions from source positives to positive expansion. Lost and new count changed rank-one successes on the identical query pool.

differentiably, the new normalizer equals the original normalizer as a function of scores. Differentiating that equality restores the original individual negative derivatives. The allocation comparison therefore requires detachment.

The transpose need not preserve the match. Row-specific matching does not generally preserve the normalizers of the transposed text-to-image objective. For example, take two images, three captions, all scores zero, positive pairs (1, 1) and (2, 2), and weights

$$W = \begin{pmatrix} 1 & 0.2 & 0.8 \\ 0.6 & 1 & 0.4 \end{pmatrix}. \quad (50)$$

Both row constants are 0.5 and both image-row losses remain $\log 2$. The two eligible text-column normalizers change from 1.6, 1.2 to 1.5, 1.5. Consequently the full symmetric loss increases by

$$\frac{1}{4} \log \frac{1.5^2}{1.6 \cdot 1.2} \approx 0.0396513. \quad (51)$$

A symmetric matched control therefore requires additional constraints. The executed fixed constant 0.25 is not a c^* -matched experiment.

Q.4 Suppression does not control every negative group

For one positive, let all negative weights lie in $[0, 1]$. Write

$$S = \sum_j e^{s_j}, \quad D = e^{s_+} + \sum_{n \in N} w_n e^{s_n}. \quad (52)$$

The inequality $D \leq S$ implies $q_+ = e^{s_+}/D \geq p_+ = e^{s_+}/S$. Hence total negative derivative satisfies

$$\sum_{n \in N} q_n = 1 - q_+ \leq 1 - p_+. \quad (53)$$

This bound does not hold for every subset of negatives. Consider exponential masses 1 for the positive, 1 for a chosen negative group, and 100 for other negatives. Weight the chosen group by 0.5

Policy	I→T R1	R5	R10	T→I R1	R5	R10
Frozen	56.32	79.82	87.12	39.35	65.40	75.62
Source	55.85	79.59	86.93	39.38	65.67	75.91
Expanded	47.29	71.56	80.89	34.84	60.77	71.50
Contradiction	47.25	71.61	80.87	34.81	60.78	71.46
Neutral	48.01	72.58	81.79	35.39	61.56	72.19
Combined	47.26	71.93	81.33	35.27	61.36	72.05
Random	43.43	67.84	77.81	29.95	55.28	66.61
Smoothing	53.85	77.94	85.71	37.08	63.47	73.72
Ranking	56.32	79.82	87.12	39.35	65.40	75.62
Fixed	54.33	77.83	85.75	36.86	62.66	73.19
Shuffled	52.27	76.63	84.83	38.19	64.41	74.58
Median	53.97	77.93	85.83	39.21	65.42	75.66
Constant	55.21	78.85	86.55	39.39	65.62	75.85

Table 61: Full-pool COCO recall at ranks one, five, and ten, averaged across selected seeds (percent).

Input	Probabilities	Acc.	Macro-F1	ECE	Brier	NLL
Image	uncalibrated	35.64	34.06	4.73	0.674	1.111
Image	calibrated	35.64	34.06	0.05	0.666	1.098
Text	uncalibrated	61.76	61.38	5.78	0.509	0.879
Text	calibrated	61.76	61.38	0.68	0.503	0.858
Multimodal	uncalibrated	65.69	64.96	12.53	0.487	0.887
Multimodal	calibrated	65.69	64.96	1.38	0.457	0.783

Table 62: Held-out relation classification on the newly sampled test set. Accuracy, macro-F1, and ECE are percentages; Brier and NLL retain their native scale.

and the others by 0.001. The chosen group’s probability changes from

$$\frac{1}{102} \approx 0.009804 \quad \text{to} \quad \frac{0.5}{1 + 0.5 + 0.1} = 0.3125. \quad (54)$$

The total negative mass decreases, but that group’s mass increases. Contradiction weighting by two also lies outside the premise $w_n \leq 1$.

Q.5 Summed-positive probability

Let $A = \sum_{p \in P} e^{s_p}$. A different objective is

$$\ell_{\text{set}} = \log D - \log A. \quad (55)$$

For $p \in P$, direct differentiation gives

$$\frac{\partial \ell_{\text{set}}}{\partial s_p} = q_p - \frac{e^{s_p}}{A} \leq 0, \quad (56)$$

because $D \geq A$. The attraction is allocated according to the positive scores, rather than equally across positives. This permits the highest-scoring positives to dominate the target distribution. The identity describes a distinct objective, not an equivalent implementation of Equation 44.

Q.6 Parameter-dependent weights and score updates

Let J_+ be a fixed nonzero-support set containing P . For $j \in J_+$, assume that $w_j(\theta) > 0$ is locally differentiable, with $w_p(\theta) \equiv 1$ on positives. All other weights are fixed at zero. With fixed targets and differentiable parameter-dependent scores,

$$\begin{aligned} \nabla_{\theta} \ell = & \sum_{j \in J_+} (q_j - y_j) \nabla_{\theta} s_j \\ & + \sum_{j \in J_+} q_j \nabla_{\theta} \log w_j. \end{aligned} \quad (57)$$

The second term vanishes for detached weights. No logarithmic derivative is taken at a zero weight; the identity does not describe a change of support through zero. Even with detached weights, a parameter update combines the score Jacobians across every candidate and batch. A sign or magnitude bound for an individual score derivative is not a corresponding bound on the parameter-gradient norm or on the score after optimization.

Input	Gate	Accept	Correct	Coverage	Precision	95% interval
Image	supported	0	0	0.00	n/a	n/a
Image	contradicted	0	0	0.00	n/a	n/a
Text	supported	0	0	0.00	n/a	n/a
Text	contradicted	0	0	0.00	n/a	n/a
Multimodal	supported	0	0	0.00	n/a	n/a
Multimodal	contradicted	683	647	11.65	94.73	[92.72, 96.59]

Table 63: Selective test predictions. Precision is undefined when no pair is accepted. Intervals resample whole test images and are descriptive.

Input	C	Temp.	Max. iter.	OOF F1
Image	0.1	3.085	1182	33.17
Text	1	1.312	2318	63.46
Multimodal	0.01	1.904	2505	67.22

Table 64: Classifier selection and completion. Maximum iteration count is over the full-training grid; OOF is image-disjoint out-of-fold macro-F1 (percent).

Q.7 Other fixed-score identities

The following facts collect the single-positive special cases behind similarity suppression. Let $0 \leq w_n \leq 1$ for negatives, $w_+ = 1$, and

$$R = \sum_{n \in N} (1 - w_n) e^{s_n}, \quad D = S - R. \quad (58)$$

Since $0 \leq R < S$,

$$\log q_+ - \log p_+ = -\log(1 - R/S) \geq R/S \geq 0. \quad (59)$$

The logarithm inequality follows by integrating $1/(1-x) \geq 1$ over $[0, R/S]$. It compares probabilities at fixed scores and does not change their raw ranking. For a group of m negatives at common score a and common weight w , its gradient mass is

$$G = \frac{mwe^a}{D}. \quad (60)$$

The denominator also depends on the group. Let A be the unweighted exponential mass outside that group, so $S = A + me^a$. Nonnegative suppression elsewhere gives

$$R \geq m(1-w)e^a, \quad \frac{R}{S} \geq \frac{m(1-w)e^a}{A + me^a}. \quad (61)$$

This rational lower bound tends to $1-w$ as m grows with A, a, w fixed. The actual log-probability gain need not saturate; with $w = 0$ and no other suppressed candidates, it can grow without bound as m increases. At equal scores, a smaller weight gives

smaller normalized probability within the same denominator. This says nothing about whether the weighting identifies a correct semantic relation.

As every weight tends to one, the weighted loss converges to the corresponding unweighted uniform-positive loss by continuity. A common negative weight below one does not yield the ordinary loss while the positive weight remains one. Finally, weighted normalization is ordinary softmax normalization on shifted scores $s_j + \log w_j$, with $\log 0 = -\infty$. It is not label smoothing, which changes the targets.

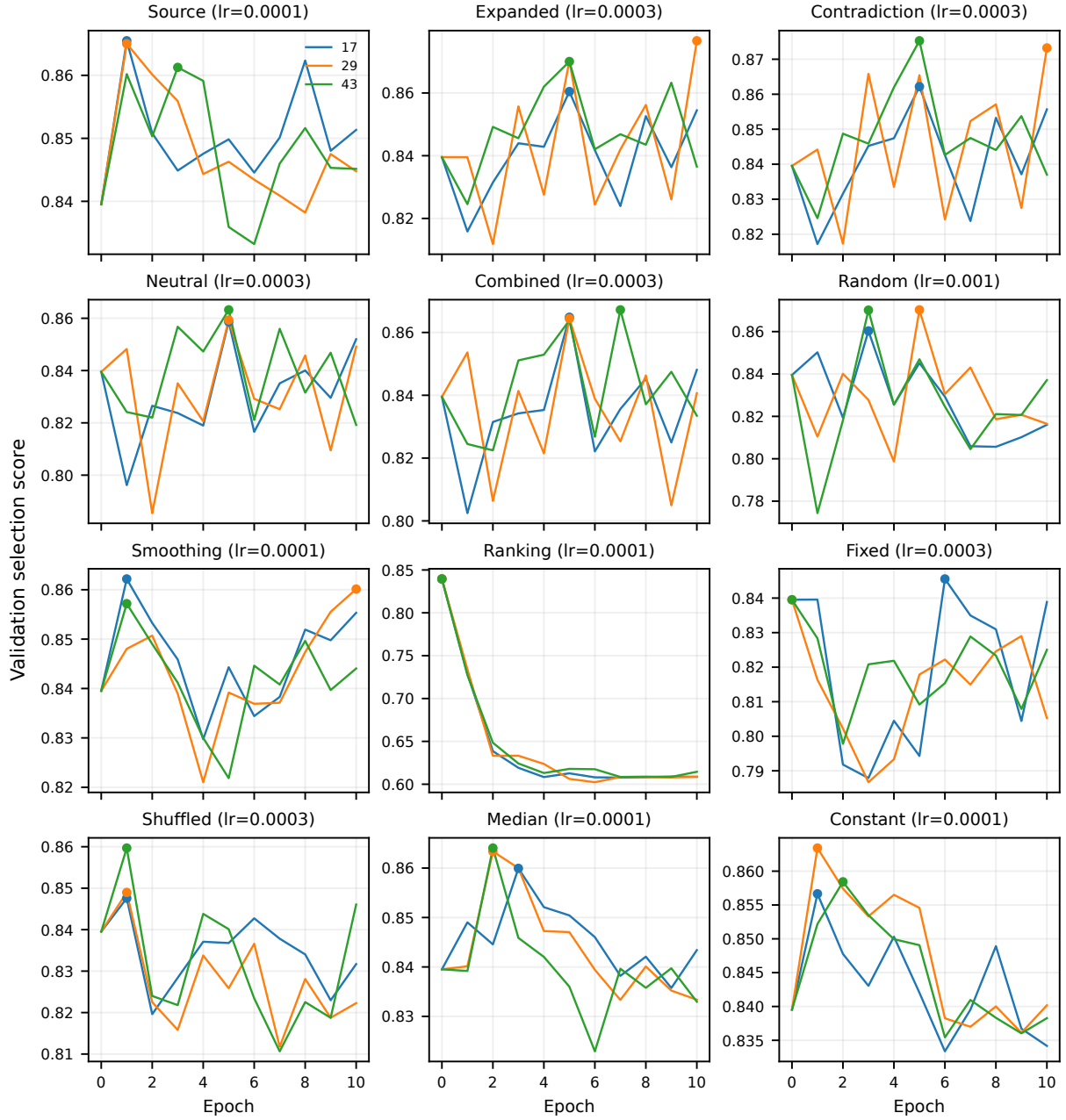


Figure 6: Validation trajectories at the selected learning rate of each policy. Lines are seeds; dots mark retained epochs. Every candidate evaluated epoch zero through ten. These are selection data, not held-out outcomes.